

MILLER INDICES OF DIRECTIONS

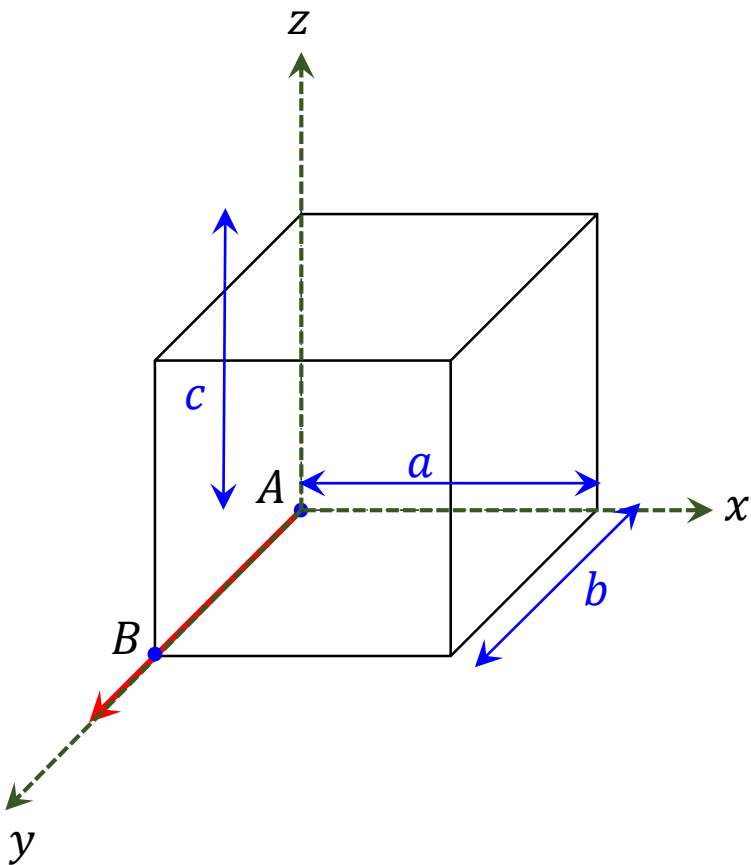
- Miller indices are standard method to identify/designate crystallographic direction and plane

Steps to find out Miller indices of directions

1. Choose a point **on the direction** as origin (*In crystallographic coordinate we are free to choose the origin anywhere we want*)
2. Choose a coordinate system with axes parallel to the unit cell edges.
3. Find the coordinate of **another point in the direction** in terms of lattice parameters in **terms of lattice parameters** (eg: $1a, 0b, -1c$)
4. Reduce the coordinates to smallest integers (either by dividing or multiplying by a common factor)
5. Put it in square bracket (negative numbers are represented by a bar over the number)

Calculate the miller indices of the direction shown in the figure

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2. Choose a coordinate system with axes parallel to the unit cell edges.
3. Find the coordinate of **another point in the direction** in terms of lattice parameters in **terms of lattice parameters** (eg: $1a, 0b, -1c$)



X-direction	Y-direction	Z-direction
0a	1b	0c

[0 1 0]

4. Reduce the coordinates to smallest integers (either by dividing or multiplying by a common factor)
5. Put it in square bracket (negative numbers are represented by a bar over the number)

Calculate the miller indices of the direction shown in the figure

$$[0\ 1\ 0]$$

$$[0\ \bar{1}\ 0]$$

$$[1\ 0\ 0]$$

$$[\bar{1}\ 0\ 0]$$

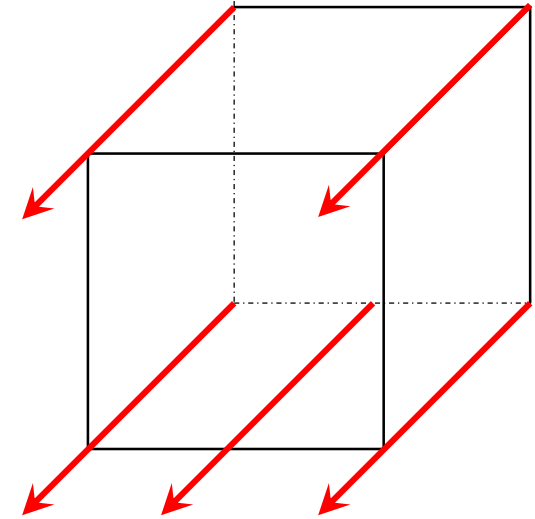
$$[0\ 0\ 1]$$

$$[0\ 0\ \bar{1}]$$

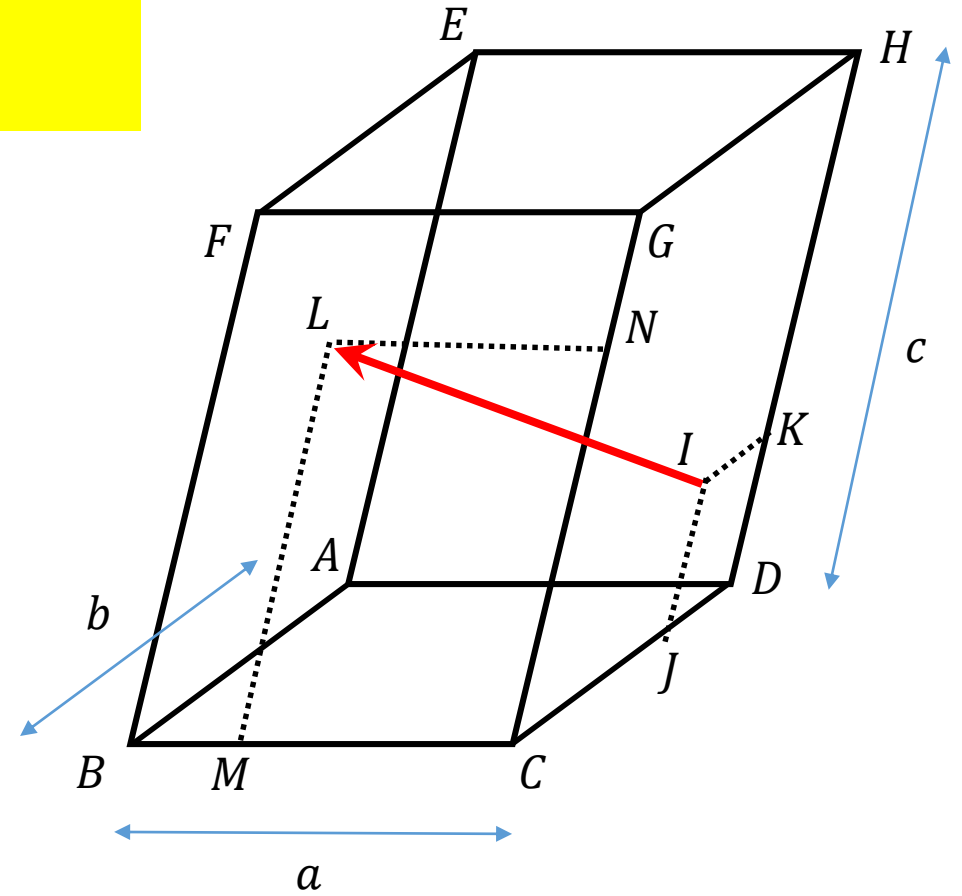
$$\langle 100 \rangle_{cubic} = [1\ 0\ 0], [0\ 1\ 0], [0\ 0\ 1], [\bar{1}\ 0\ 0], [0\ \bar{1}\ 0], [0\ 0\ \bar{1}]$$

$$\langle 100 \rangle_{tetragonal} = [1\ 0\ 0], [0\ 1\ 0], [\bar{1}\ 0\ 0], [0\ \bar{1}\ 0]$$

$$\langle 100 \rangle_{orthogonal} = [1\ 0\ 0], [\bar{1}\ 0\ 0]$$

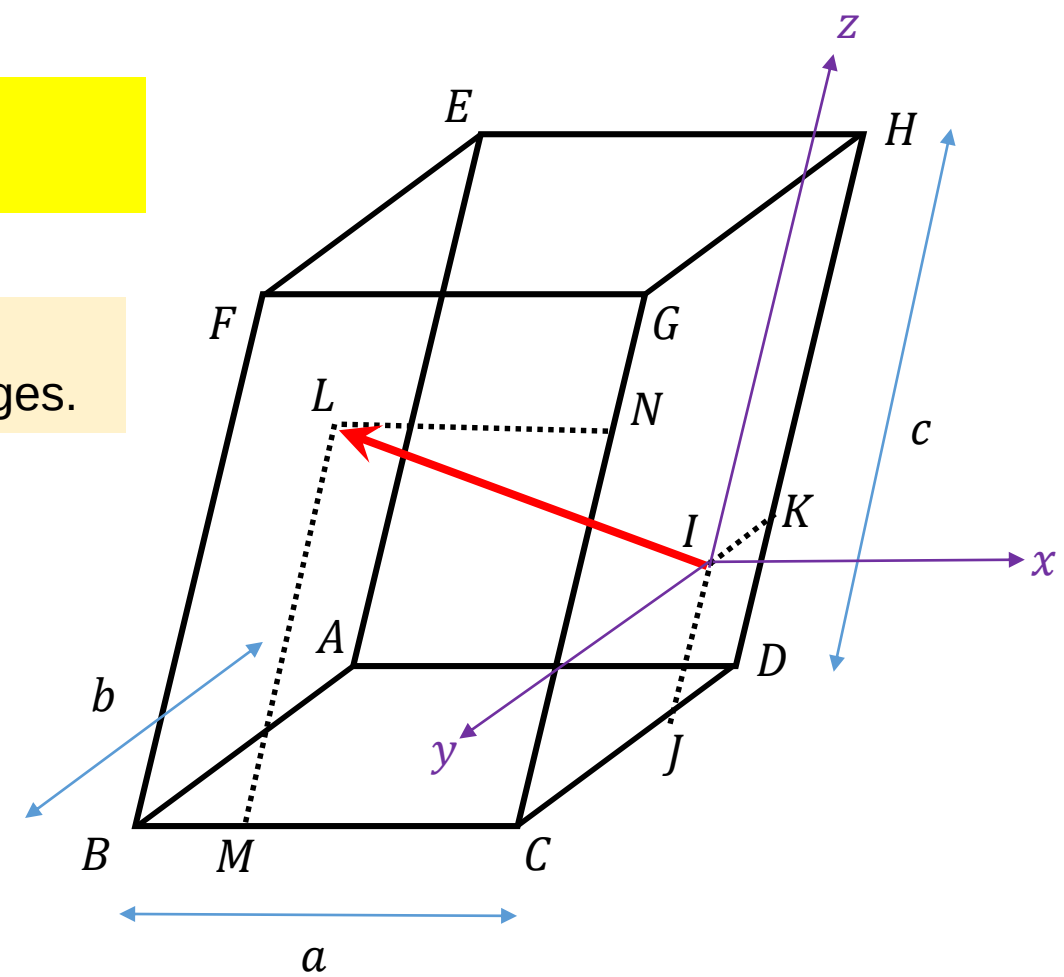


Calculate the miller indices of the direction IL
Given $a=10$, $b=8$, $c=12$, $IK=2$, $IJ=4$, $LM=9$, $LN=7$



Calculate the miller indices of the direction IL
Given $a=10$, $b=8$, $c=12$, $IK=2$, $IJ=4$, $LM=9$, $LN=7$

Choose a point **on the direction** as origin
Choose a coordinate system with axes parallel to the unit cell edges.

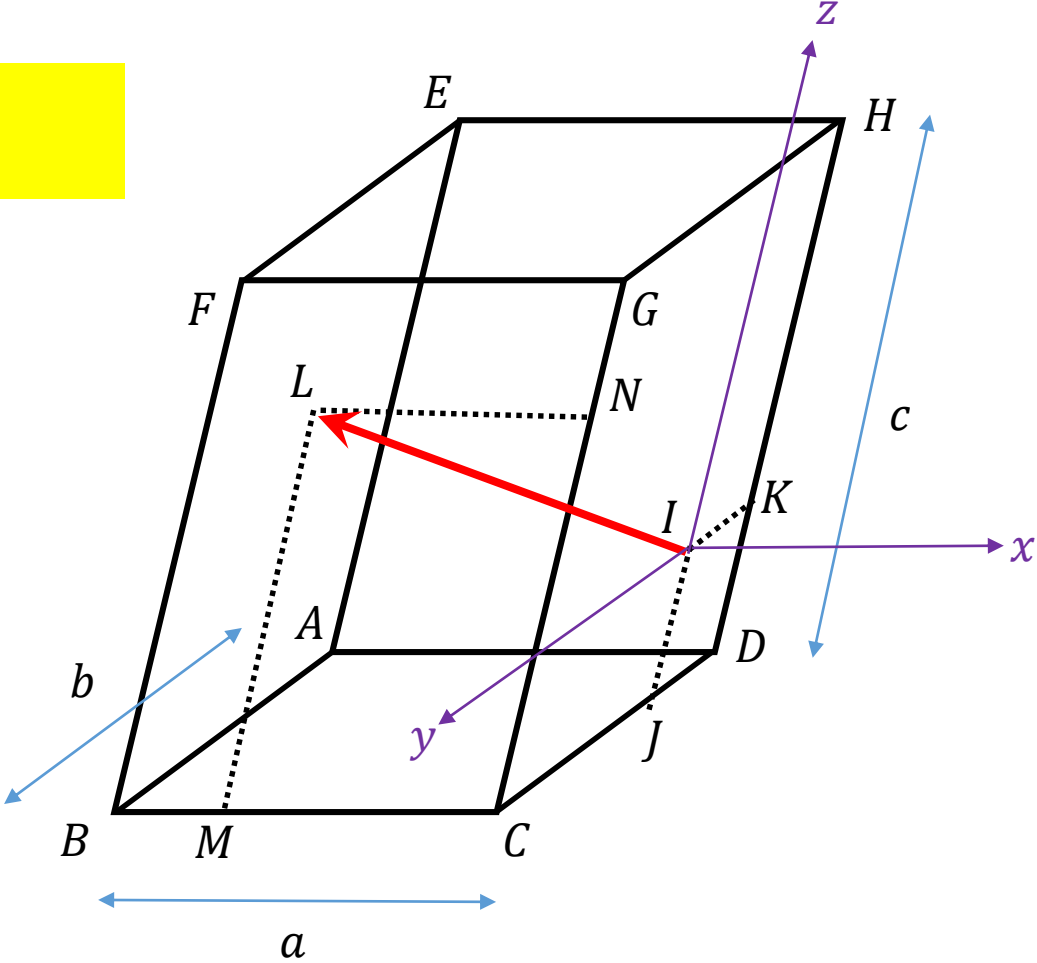


Calculate the miller indices of the direction IL
 Given a=10, b=8, c=12, IK=2, IJ=4, LM=9, LN=7

- Find the coordinate of **another point in the direction** in terms of lattice parameters in **terms of lattice parameters** (eg: 1a, 0b, -1c)

X-direction	Y-direction	Z-direction
CM (-) (CM=LN=7)	CJ (+) (CD-IK=8-2=6)	IJ(-), ML(+) IJ=4 ML=LM=9
-7	6	9-4=5
$-\frac{7}{10}a$	$\frac{6}{8}b$	$\frac{5}{12}c$

$$-\frac{7}{10} \quad \frac{6}{8} \quad \frac{5}{12}$$



Calculate the miller indices of the direction IL
 Given $a=10$, $b=8$, $c=12$, $IK=2$, $IJ=4$, $LM=9$, $LN=7$

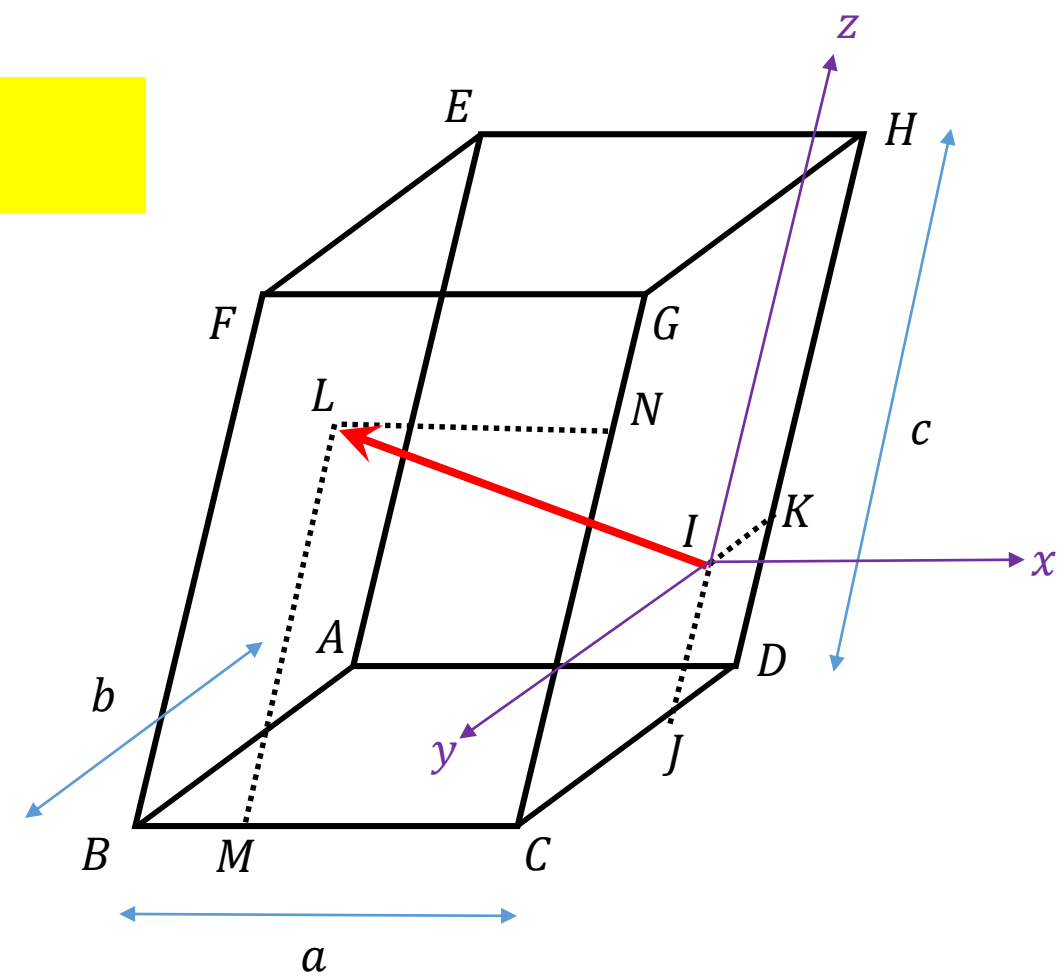
$$-\frac{7}{10} \quad \frac{6}{8} \quad \frac{5}{12}$$

Reduce the coordinates to smallest integers (either by dividing or multiplying by a common factor)

Put it in square bracket (negative numbers are represented by a bar over the number)

$$-42 \ 45 \ 25$$

$$[\bar{42} \ 45 \ 25]$$

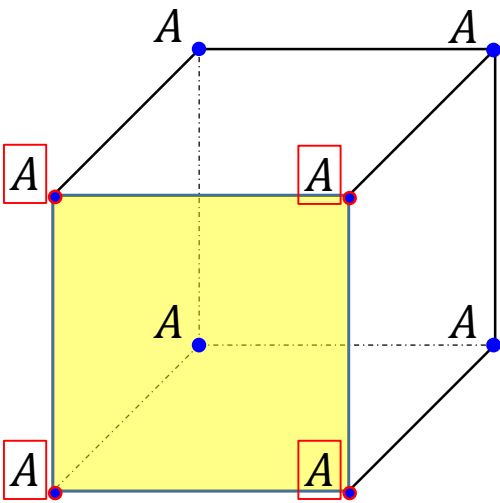


MILLER INDICES OF PLANES

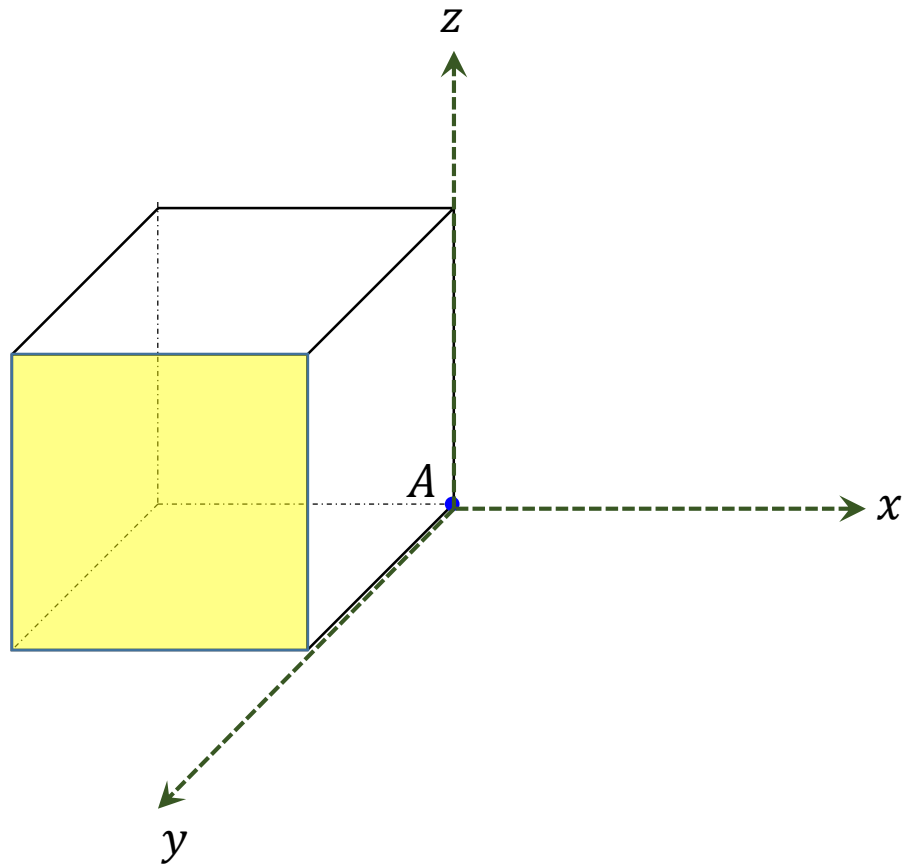
Steps to find out Miller indices of Planes

1. Choose a point **not on the plane** as origin (*In crystallographic coordinate we are free to choose the origin anywhere we want*)
2. Choose a coordinate system with axes parallel to the unit cell edges.
3. Find the **intercepts** along axes in terms of **lattice parameters**
- 4. Take Reciprocals**
5. Reduce the reciprocals to smallest integers (either by dividing or multiplying by a common factor)
6. Enclose in parenthesis (negative numbers are represented by a bar over the number)

Calculate the miller indices of the Plane shown in the figure

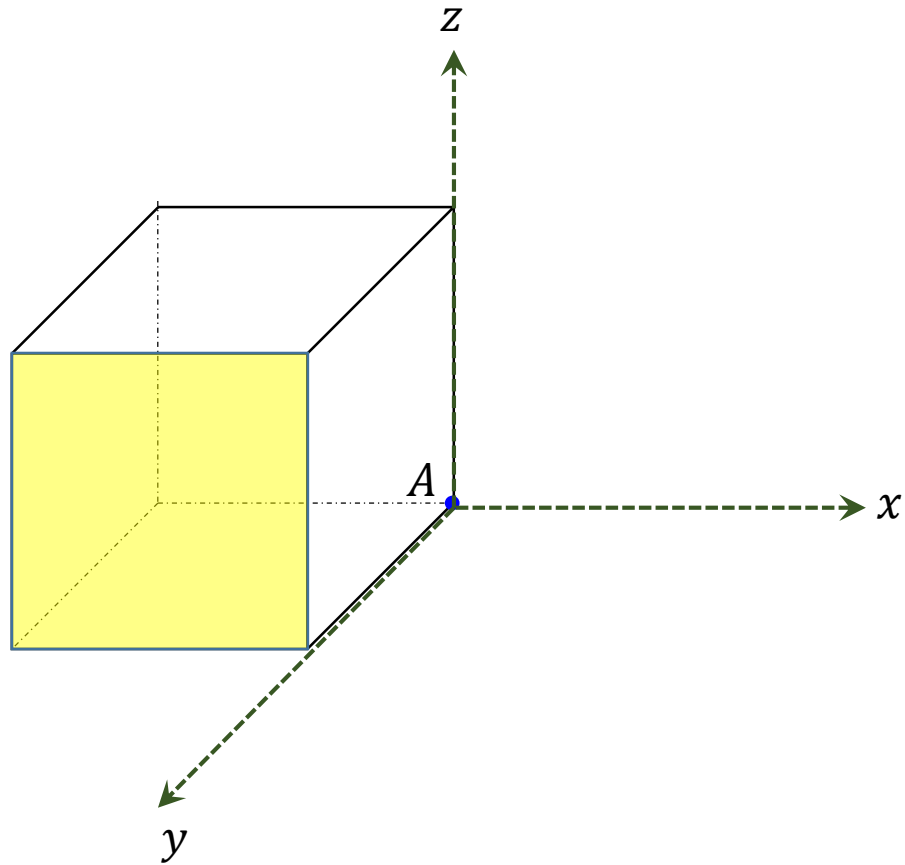


Calculate the miller indices of the Plane shown in the figure



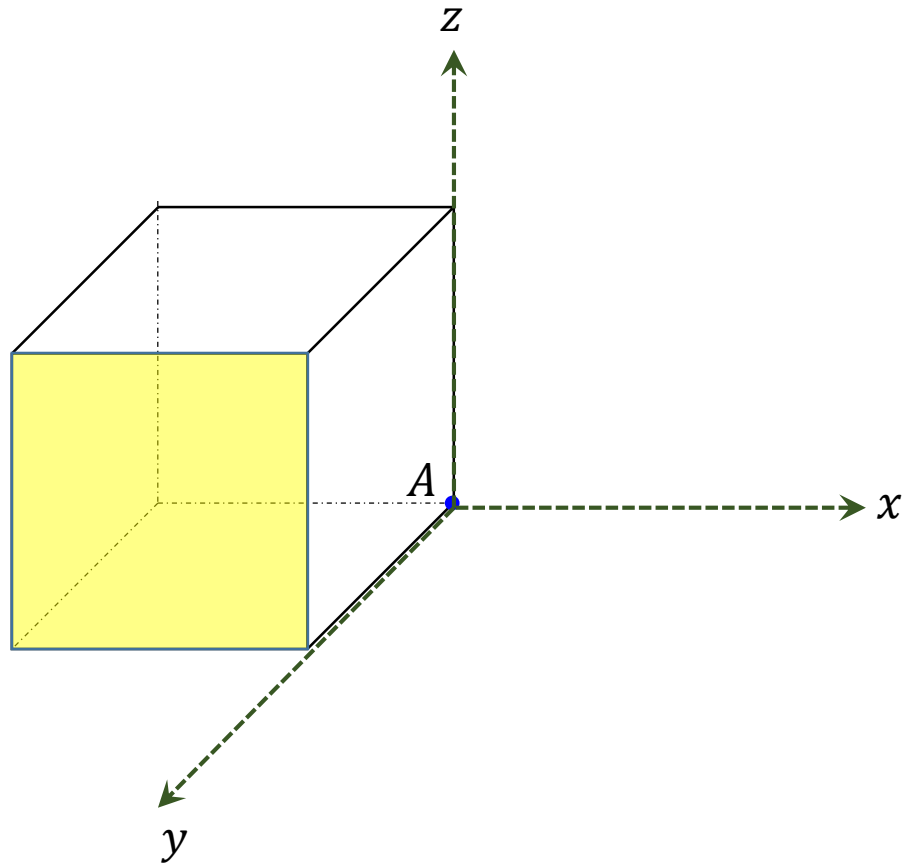
Calculate the miller indices of the Plane shown in the figure

Find the **intercepts** along axes in terms of **lattice parameters**



X-direction	Y-direction	Z-direction
α	$1b$	α

Calculate the miller indices of the Plane shown in the figure



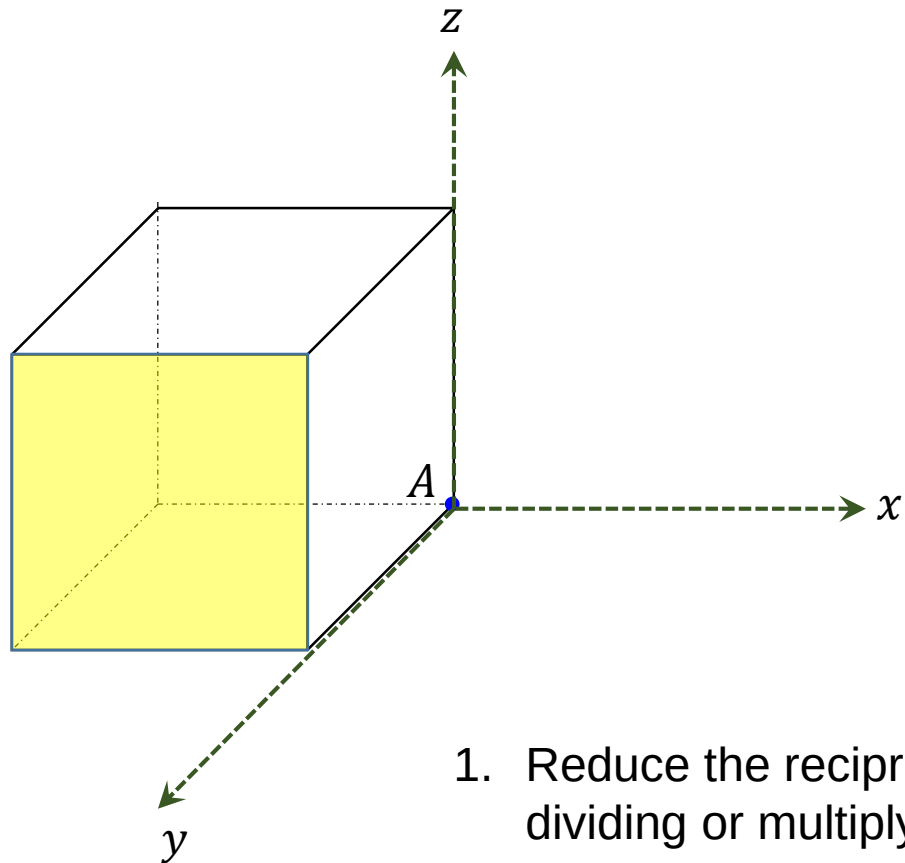
Find the **intercepts** along axes in terms of **lattice parameters**

X-direction	Y-direction	Z-direction
∞	$1b$	∞
∞	1	∞

Take Reciprocals

X-direction	Y-direction	Z-direction
0	1	0

Calculate the miller indices of the Plane shown in the figure



Find the **intercepts** along axes in terms of **lattice parameters**

X-direction	Y-direction	Z-direction
∞	$1b$	∞
∞	1	∞

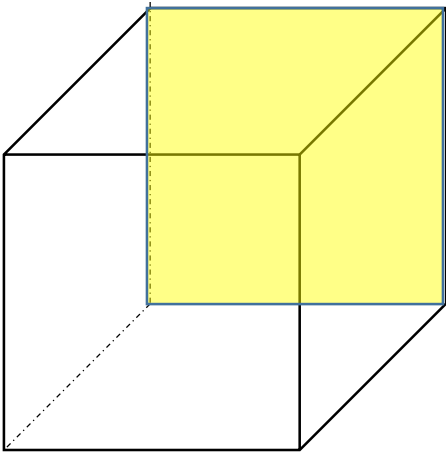
Take Reciprocals

X-direction	Y-direction	Z-direction
0	1	0

1. Reduce the reciprocals to smallest integers (either by dividing or multiplying by a common factor)
2. Enclose in parenthesis (negative numbers are represented by a bar over the number)

$$(0 \ 1 \ 0)$$

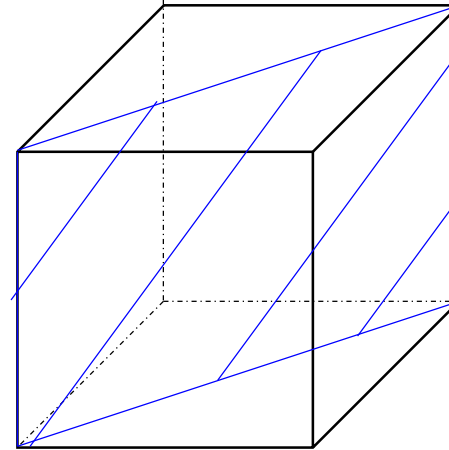
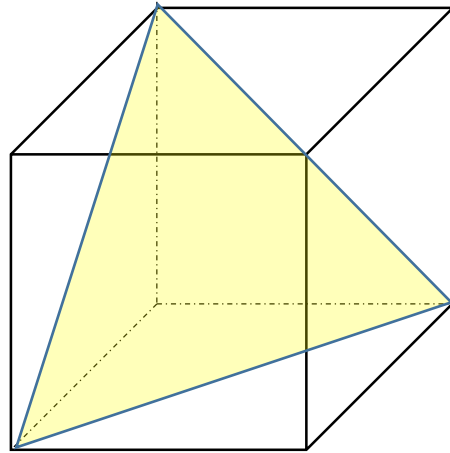
Calculate the miller indices of the Plane shown in the figure



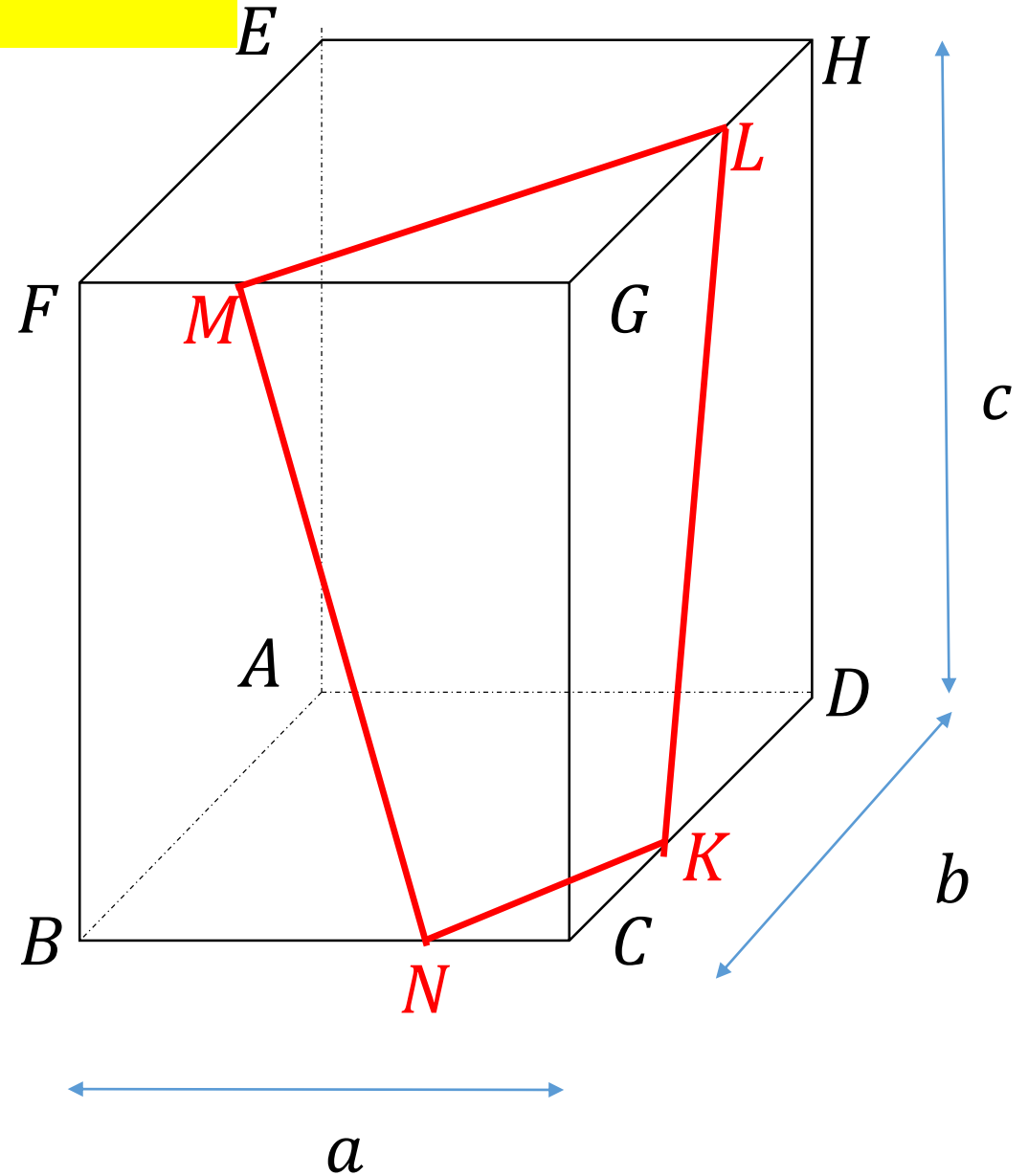
Miller Indices of directions represents only the orientation of the line corresponding to the direction and not its position or sense

Miller Indices of planes represents only the orientation of the not its position or sense

General representation $[h\ k\ l]$ or $(h\ k\ l)$



Calculate the miller indices of the Plane KLMN
Given, HL=1.5, KC=1, NC=2, FM=1, $a=8$, $b=5$, $c=10$



INTERPLANAR SPACING

d_{hkl}

The interplanar spacing between two adjacent planes having miller indices (h k l) is defined as the distance between first such plane from a parallel plane passing through origin

The inter-planar spacing in cubic/ tetragonal/
orthorhombic crystals

$$\alpha = \beta = \gamma = 90^\circ$$

INTERPLANAR SPACING

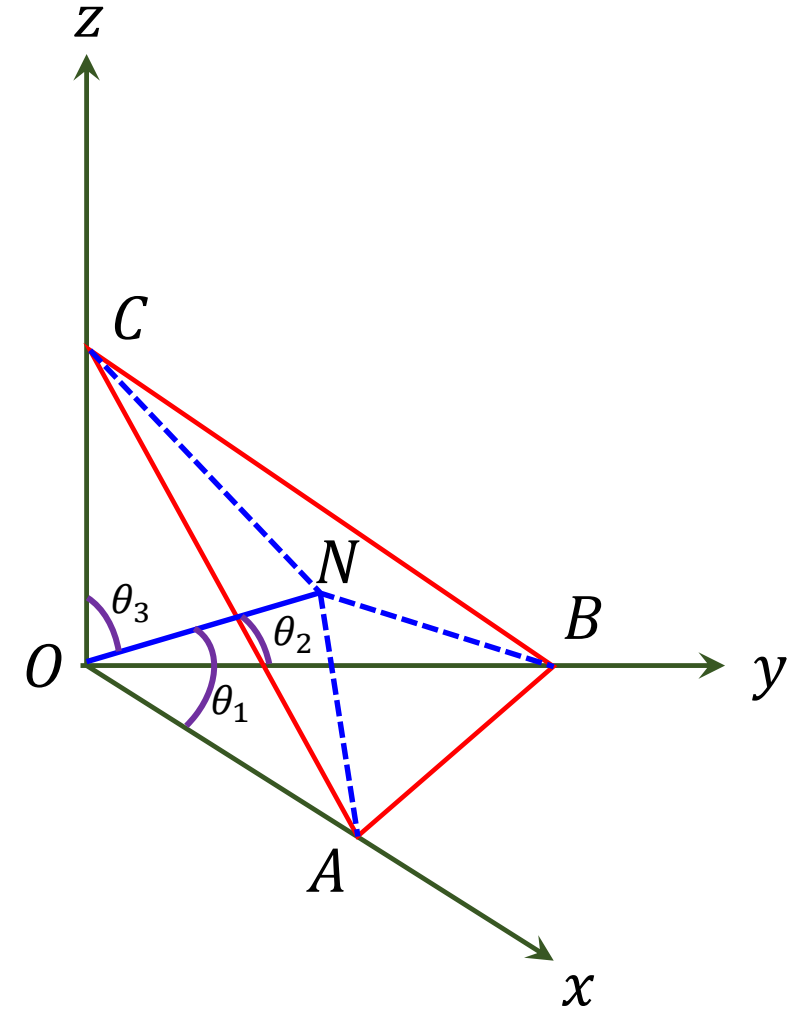
Miller indices of the plane is $(h \ k \ l)$

Co-ordinates of the intercepts $\left(\frac{a}{h}, \frac{b}{k}, \frac{c}{l}\right)$

θ_1 : angle between ON and X-axis

θ_2 : angle between ON and Y-axis

θ_3 : angle between ON and Z-axis

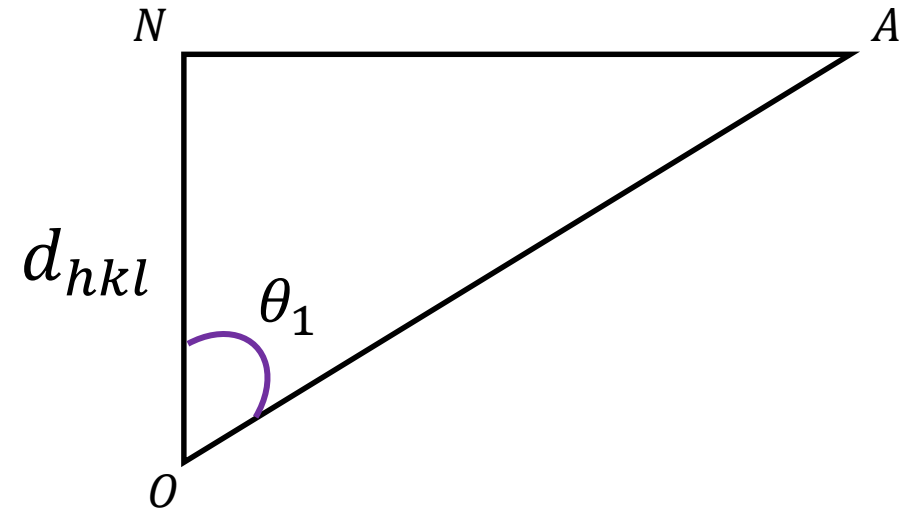


INTERPLANAR SPACING

$$\cos \theta_1 = \frac{ON}{OA} = \frac{d_{hkl}}{\left(\frac{a}{h}\right)}$$

$$\cos \theta_2 = \frac{ON}{OB} = \frac{d_{hkl}}{\left(\frac{b}{h}\right)}$$

$$\cos \theta_3 = \frac{ON}{OC} = \frac{d_{hkl}}{\left(\frac{c}{h}\right)}$$



DIRECTION COSINE

$$OP = \sqrt{(x^2 + y^2 + z^2)}$$

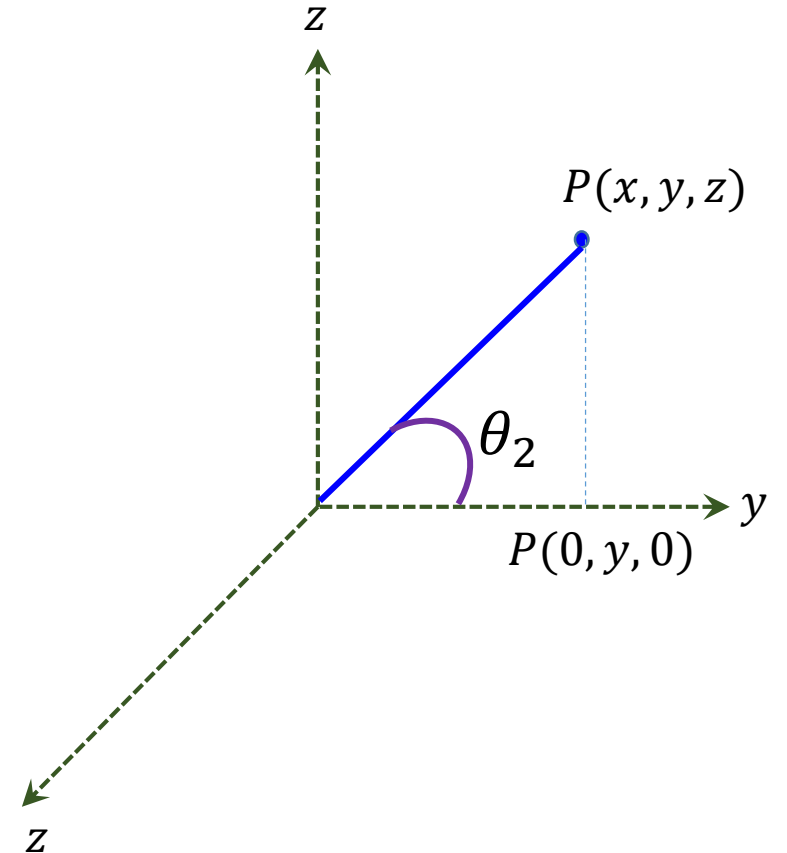
$$OY = y$$

$$\cos \theta_2 = \frac{OY}{OP} = \frac{y}{\sqrt{(x^2 + y^2 + z^2)}}$$

$$\cos \theta_1 = \frac{x}{\sqrt{(x^2 + y^2 + z^2)}}$$

$$\cos \theta_3 = \frac{z}{\sqrt{(x^2 + y^2 + z^2)}}$$

$$\cos^2 \theta_1 + \cos^2 \theta_2 + \cos^2 \theta_3 = 1$$



INTERPLANAR SPACING

$$\cos^2 \theta_1 + \cos^2 \theta_2 + \cos^2 \theta_3 = 1$$

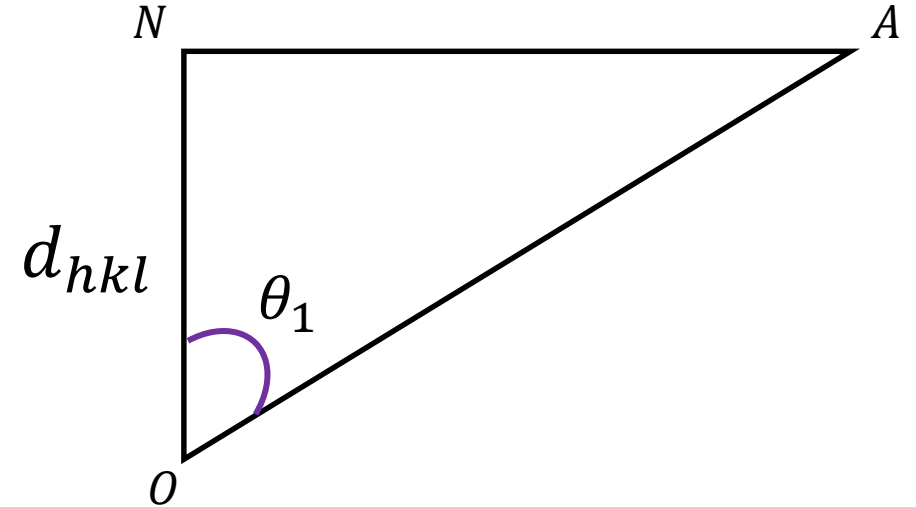
$$\cos \theta_1 = \frac{ON}{OA} = \frac{d_{hkl}}{\left(\frac{a}{h}\right)} \quad \cos \theta_2 = \frac{ON}{OB} = \frac{d_{hkl}}{\left(\frac{b}{h}\right)}$$

$$\cos \theta_3 = \frac{ON}{OC} = \frac{d_{hkl}}{\left(\frac{c}{h}\right)}$$

$$\left(\frac{d_{hkl}}{\frac{a}{h}}\right)^2 + \left(\frac{d_{hkl}}{\frac{b}{h}}\right)^2 + \left(\frac{d_{hkl}}{\frac{c}{h}}\right)^2 = 1$$

$$d_{hkl}^2 = \frac{1}{\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}}$$

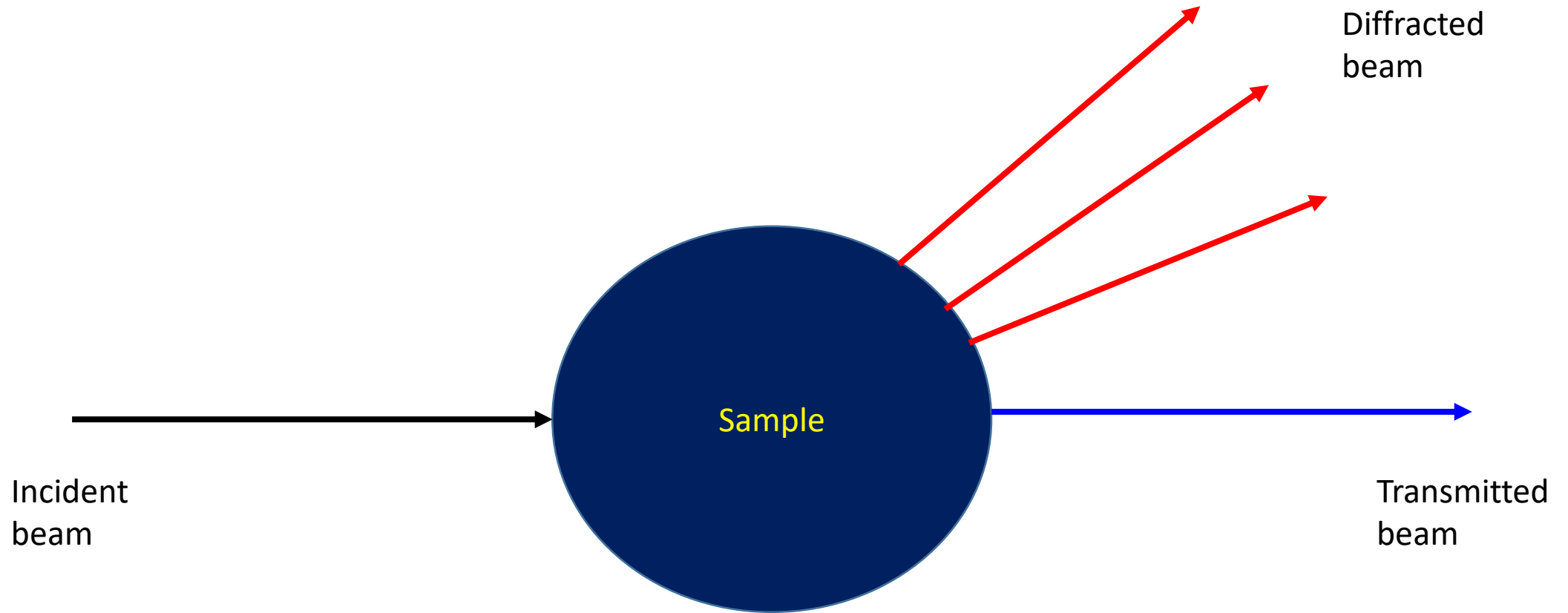
$$d_{hkl}^2 = \frac{a}{h^2 + k^2 + l^2}$$



Q. The distance between consecutive (111) planes is $2A$. *Determine the lattice parameters*

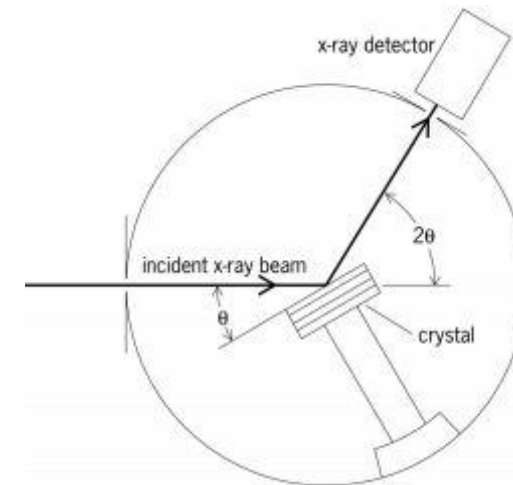
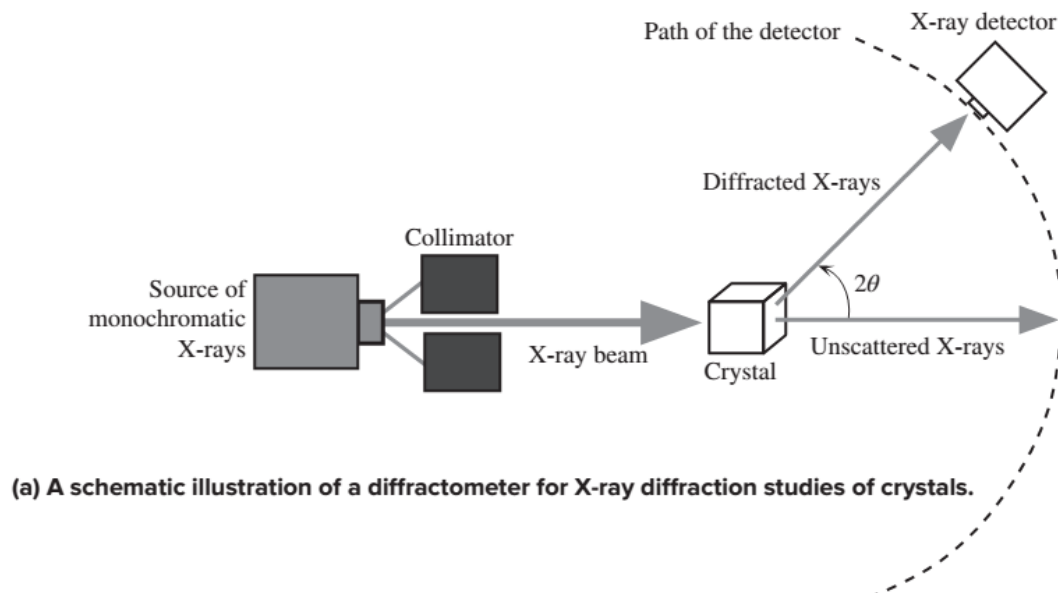
Q. In tetragonal crystal, the lattice parameters

X- RAY DIFFRACTION



These diffracted beam if analyzed properly can be used to analyze crystals

X- RAY DIFFRACTION

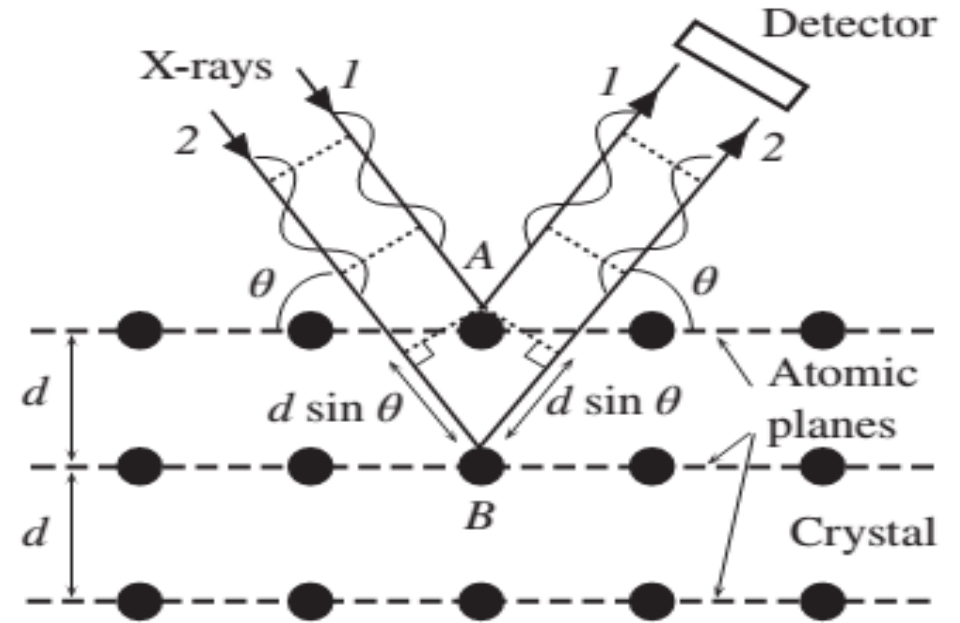


Bragg's Observation: For every diffracted beam, there exists a set of crystal lattice planes such that the diffracted beam appears to be specularly reflected from this set of planes.

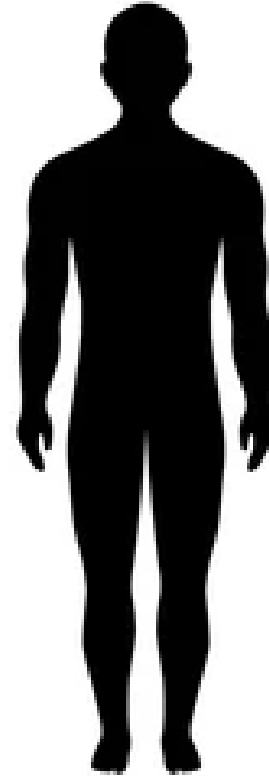
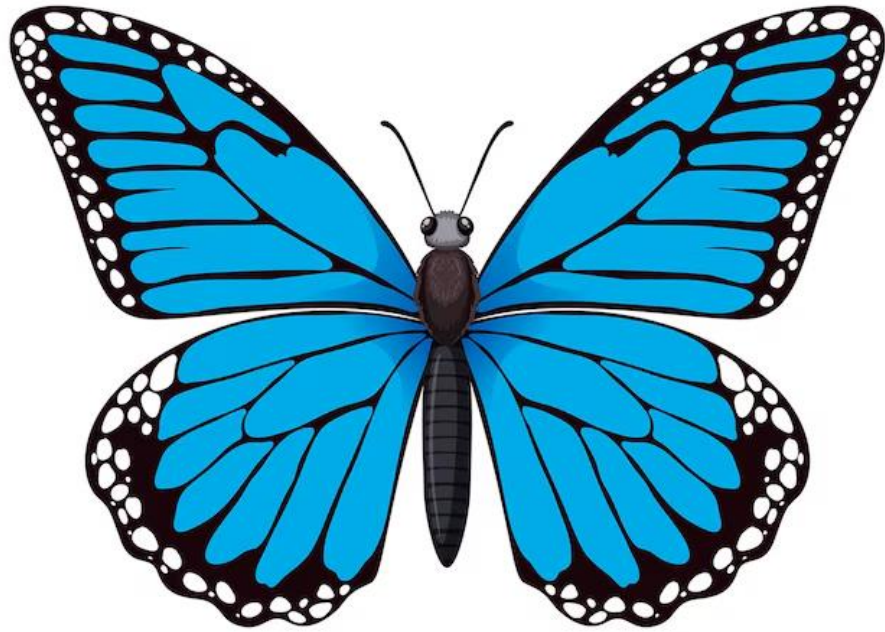
Bragg's diffraction condition

- Bragg diffraction occurs when radiation, with a wavelength comparable to atomic spacings, is scattered in a specular fashion by the atoms of a crystalline system, and undergoes constructive interference.
- When the scattered waves interfere constructively, they remain in phase since the difference between the path lengths of the two waves is equal to an integer multiple of the wavelength.
- The path difference between two waves undergoing interference is given by $2d\sin\theta$, where θ is the glancing angle.
- For constructive interference

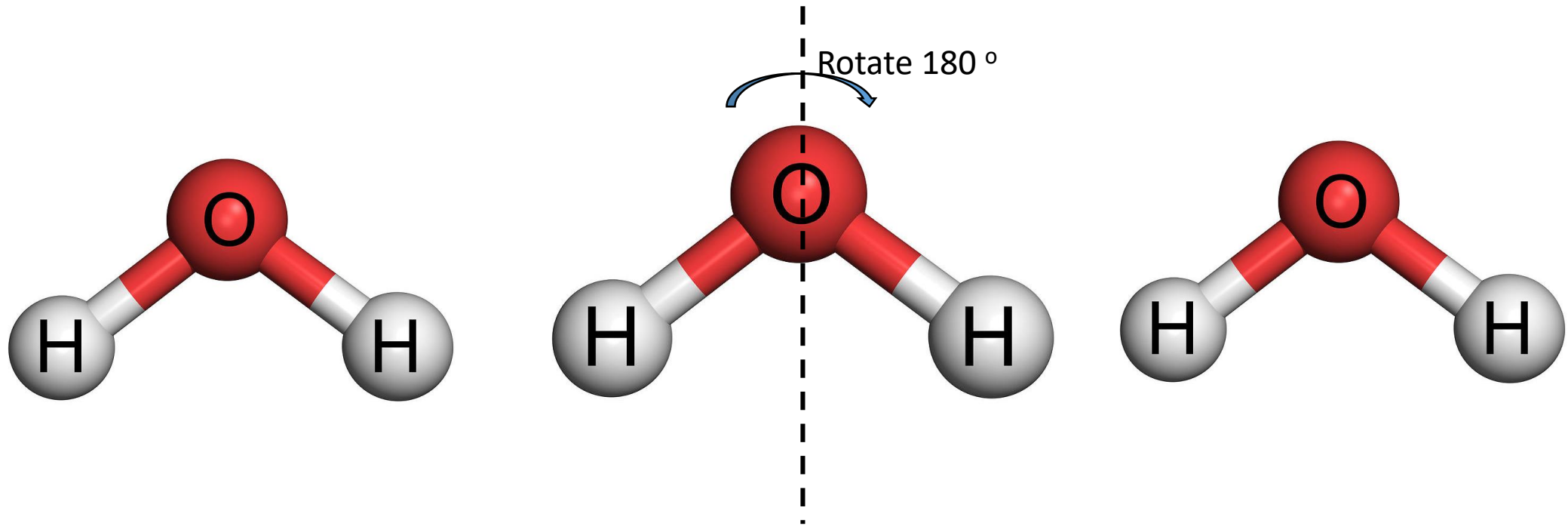
where n is a positive integer and λ is the wavelength of the incident wave. This is known as Bragg's diffraction condition.



Often we don't realize it, but we continuously live with symmetry... Symmetry is the consistency, the repetition of something in space and/or in time

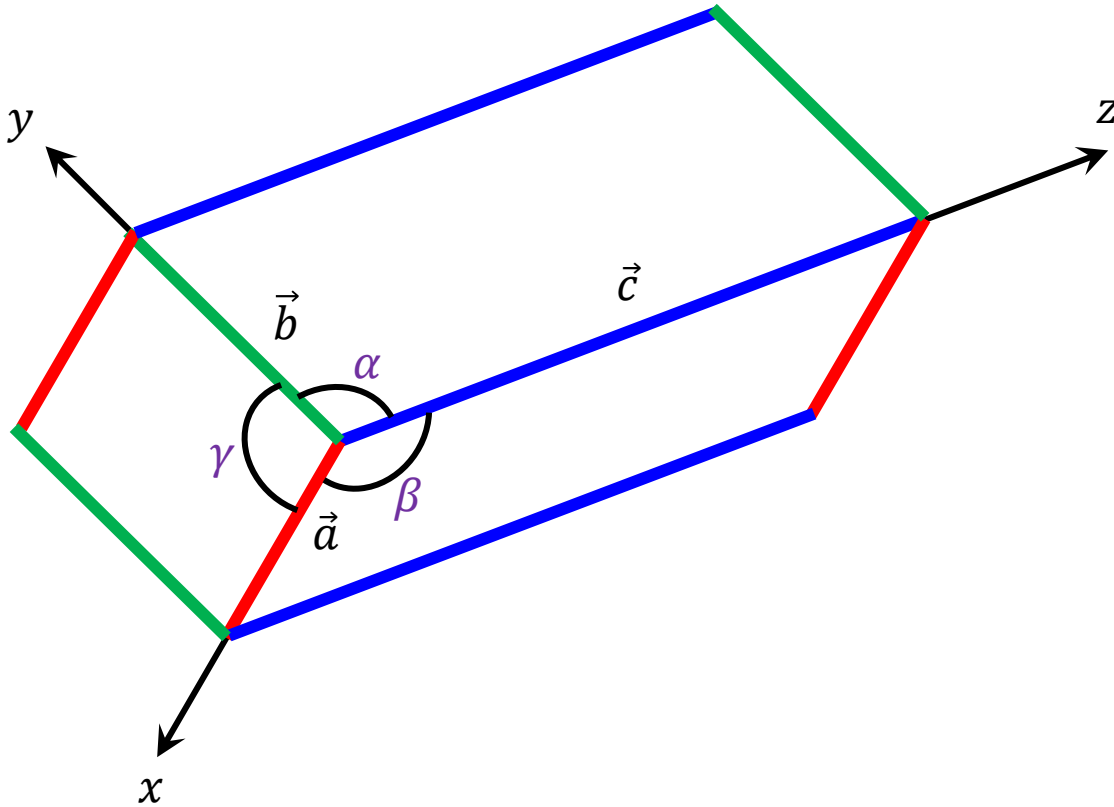


An object of figure is said to have symmetry if some movement of the figure or operation on the figure leaves it in a position indistinguishable from its original position.



Rotation is a **symmetry operation** and **rotational axis** is a **symmetry element**

LATTICE PARAMETERS



$a, b, c, \alpha, \beta, \gamma$

α : angle between $\vec{b}$ and $\vec{c}$

β : angle between $\vec{c}$ and $\vec{a}$

γ : angle between $\vec{a}$ and $\vec{b}$

CLASSIFICATION

P: Simple or primitive

F: Face Centered

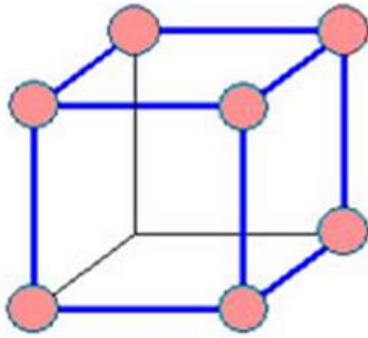
I: Body Centered

C: Base Centered or Edge Centered

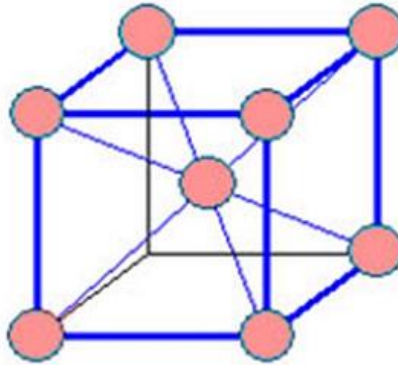
Crystal System	Conventional Unit Cell	Bravais Lattice			
Cubic	$a = b = c$ $\alpha = \beta = \gamma = 90^\circ$	P	I	F	-
Tetragonal	$a = b \neq c$ $\alpha = \beta = \gamma = 90^\circ$	P	I	-	-
Orthorhombic	$a \neq b \neq c$ $\alpha = \beta = \gamma = 90^\circ$	P	I	F	C
Monoclinic	$a \neq b \neq c$ $\alpha = \gamma = 90^\circ, \beta \neq 90^\circ$	P	-	-	C
Rhombohedral	$a = b = c$ $\alpha = \beta = \gamma \neq 90^\circ$	P			
Hexagonal	$a = b \neq c$ $\alpha = \beta = 90^\circ, \gamma = 120^\circ$	P			
Triclinic	$a \neq b \neq c$ $\alpha \neq \beta \neq \gamma$	P			

Crystal Systems

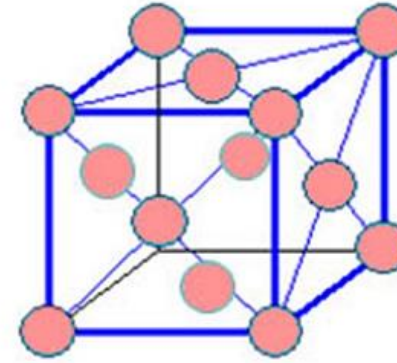
Cubic: $a = b = c$, $\alpha = \beta = \gamma = 90^\circ$



Simple
cubic

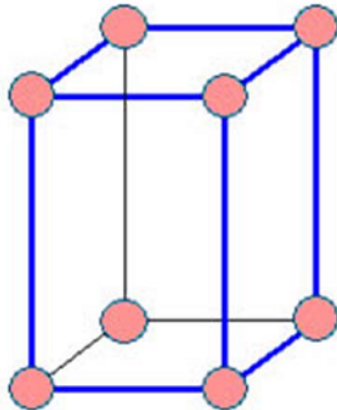


Body-centered
cubic (BCC)

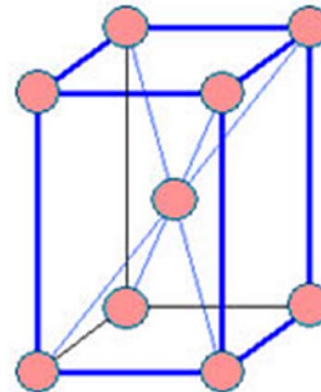


Face-centered
cubic (FCC)

Tetragonal: $a = b \neq c$, $\alpha = \beta = \gamma = 90^\circ$



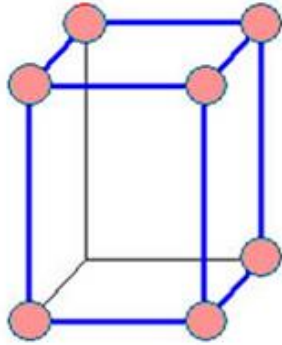
Simple
Tetragonal



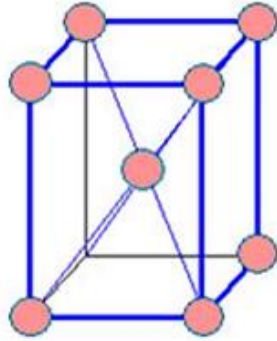
Body-centered
Tetragonal (BCT)

Crystal Systems

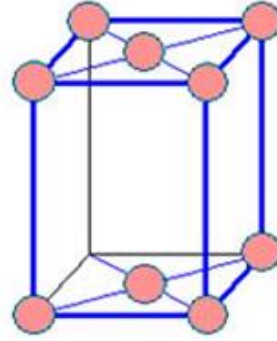
Orthorhombic: $a \neq b \neq c$, $\alpha = \beta = \gamma = 90^\circ$



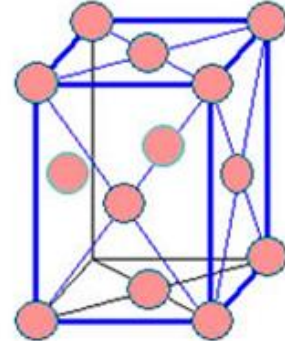
Simple



Body-centered

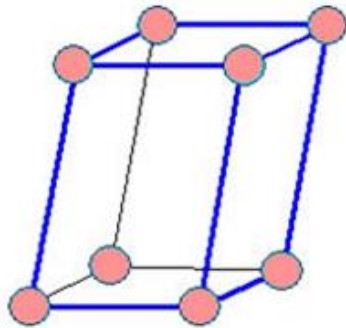


Base-centered

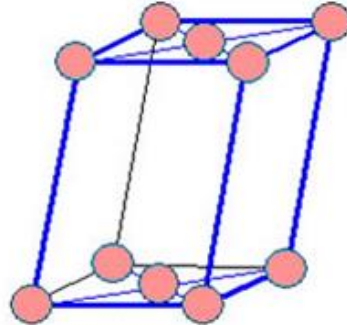


Face-centered

Monoclinic: $a \neq b \neq c$, $\alpha = \gamma = 90^\circ \neq \beta$

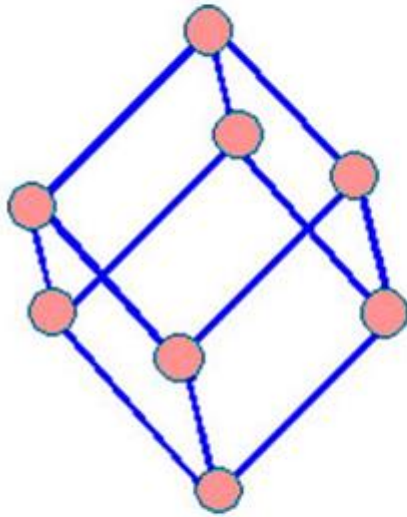


Simple
monoclinic

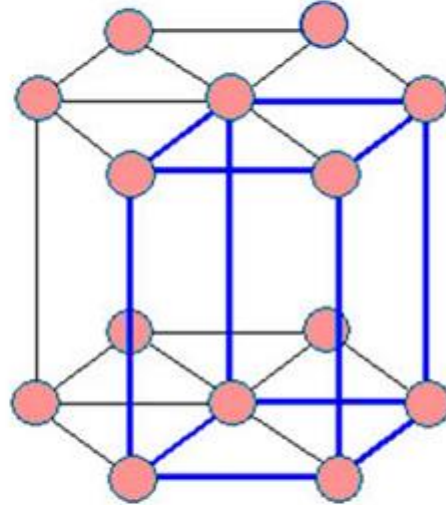


Base-centered
monoclinic

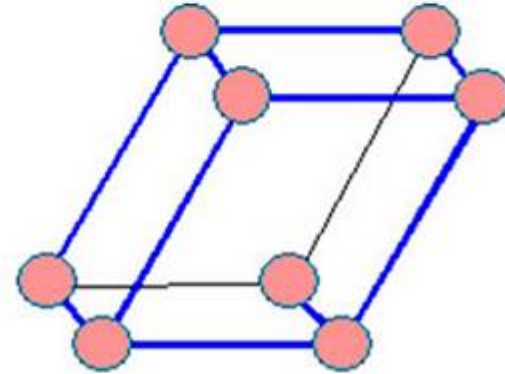
Crystal Systems



Rhombohedral
 $a = b = c$
 $\alpha = \beta = \gamma \neq 90^\circ$

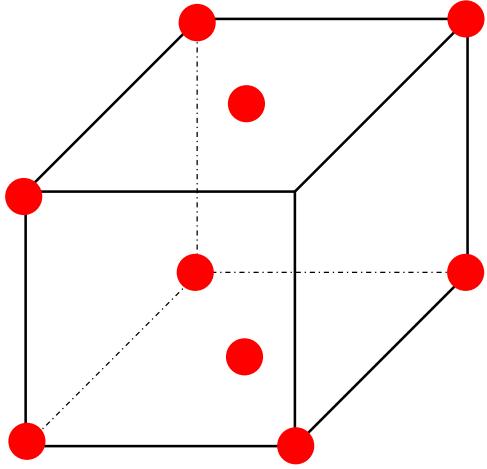


Hexagonal
 $a = b \neq c$
 $\alpha = \beta = 90^\circ \gamma = 120^\circ$

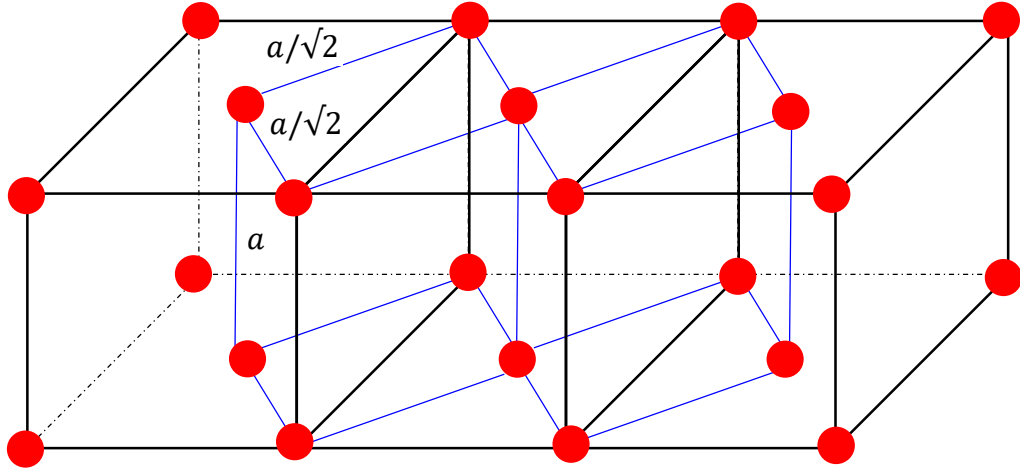


Triclinic
 $a \neq b \neq c$
 $\alpha \neq \beta \neq \gamma \neq 90^\circ$

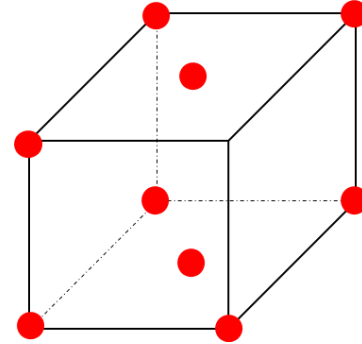
WHY NO CUBIC C?



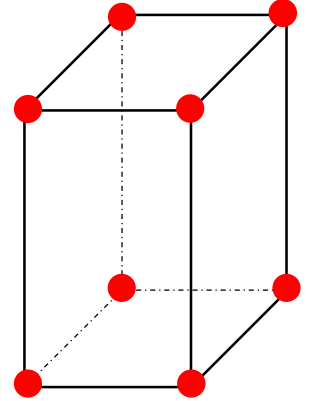
WHY NO CUBIC C?



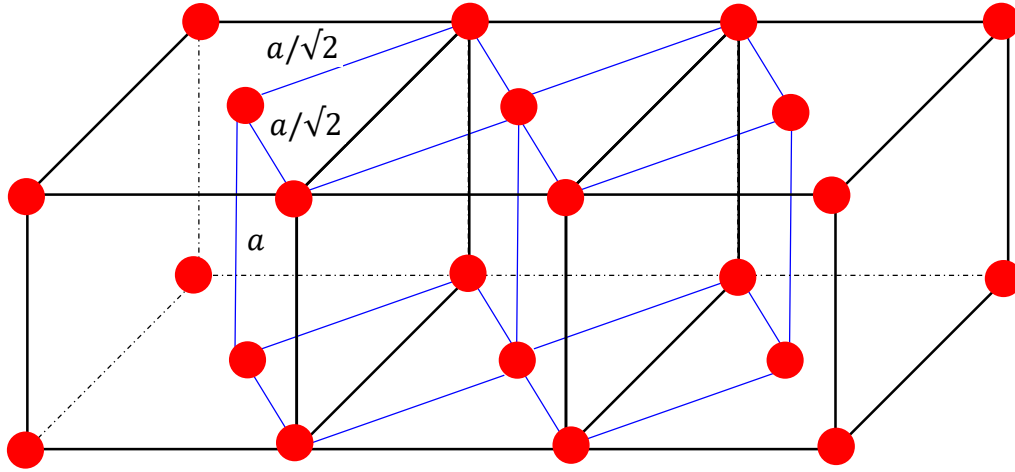
$$a = b = c$$
$$\alpha = \beta = \gamma = 90^\circ$$



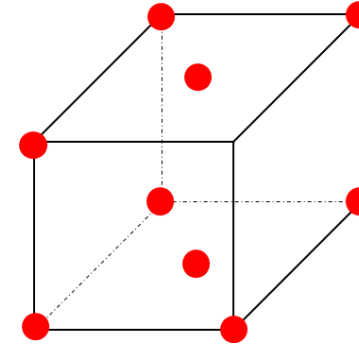
$$a = b \neq c$$
$$\alpha = \beta = \gamma = 90^\circ$$



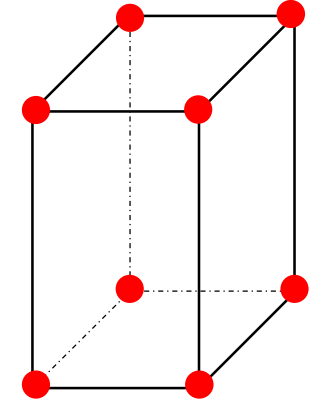
WHY NO CUBIC C?



$$a = b = c$$
$$\alpha = \beta = \gamma = 90^\circ$$

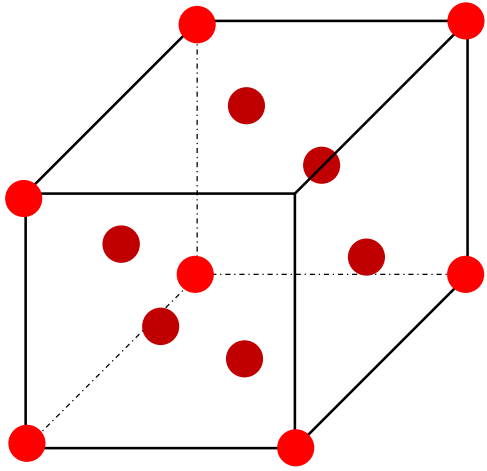


$$a = b \neq c$$
$$\alpha = \beta = \gamma = 90^\circ$$



Cubic C and **Tetragonal P** are same and so we **keep** tetragonal P and **discard** cubic C?

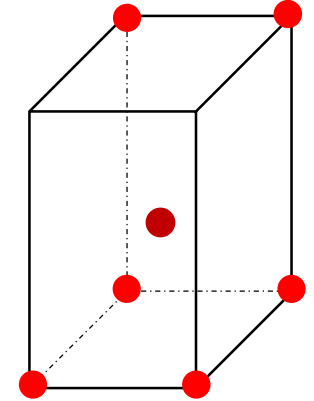
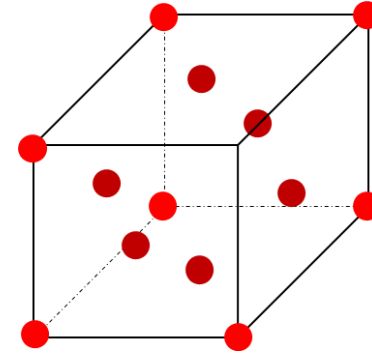
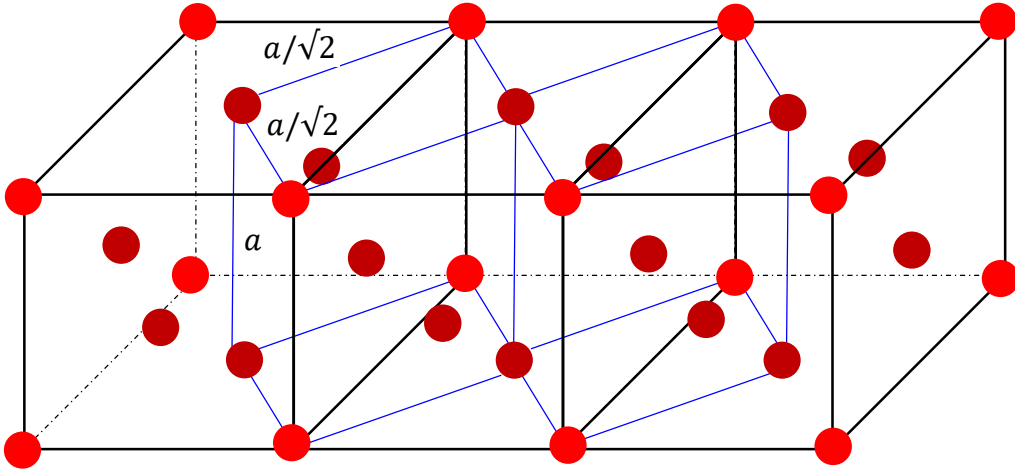
WHAT ABOUT NO CUBIC F?



WHAT ABOUT NO CUBIC F?

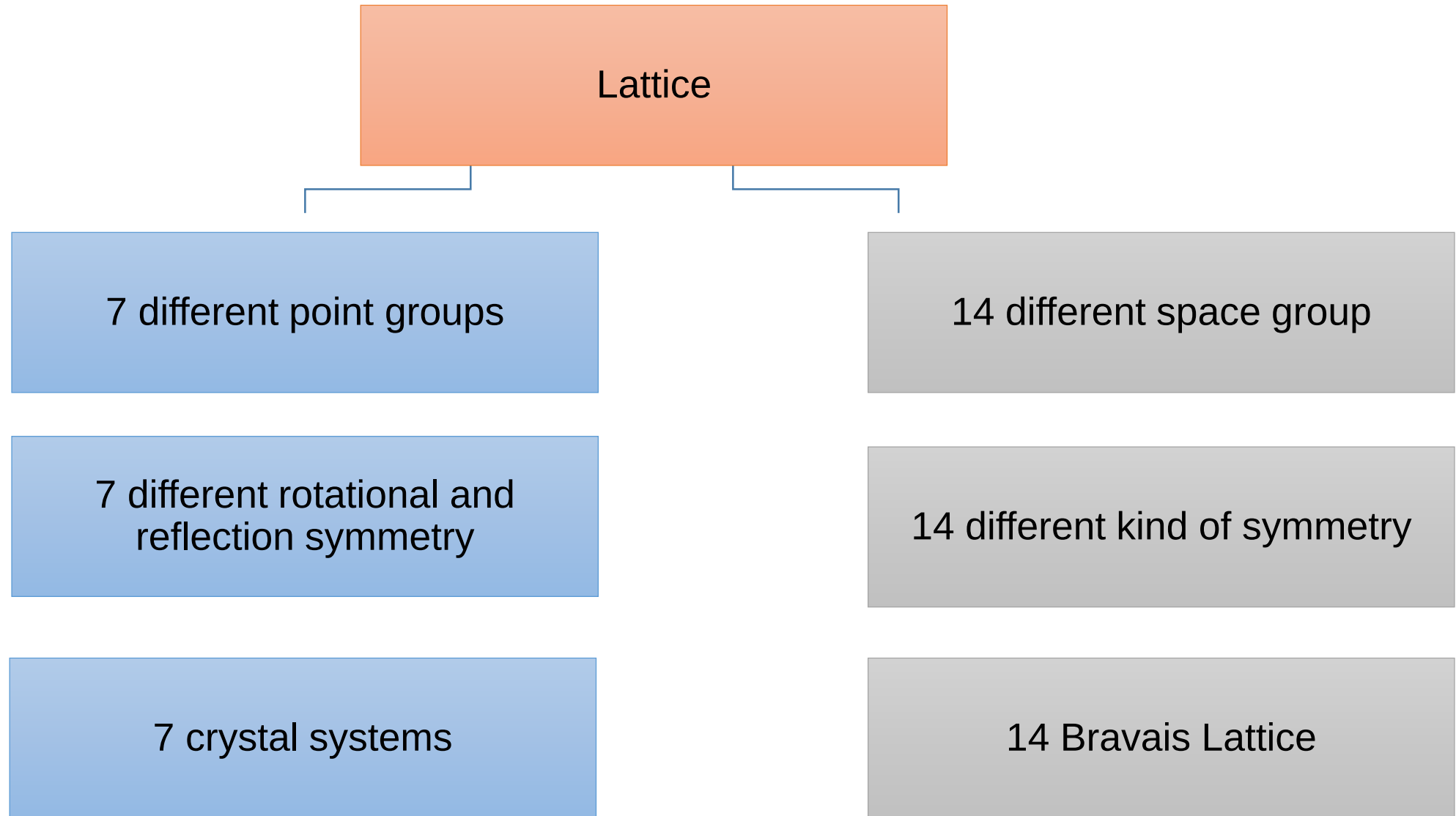
$$a = b = c$$
$$\alpha = \beta = \gamma = 90^\circ$$

$$a = b \neq c$$
$$\alpha = \beta = \gamma = 90^\circ$$



Cubic F and **Tetragonal I** are same and we **keep both** tetragonal I and cubic F?

CLASSIFICATION



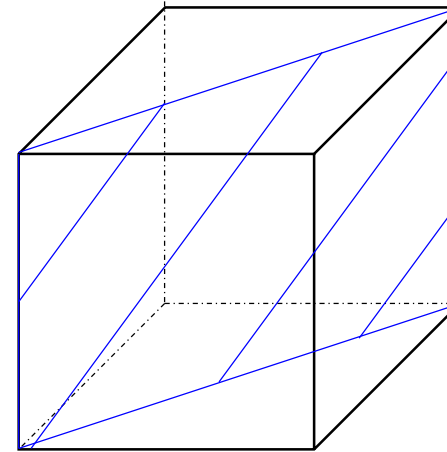
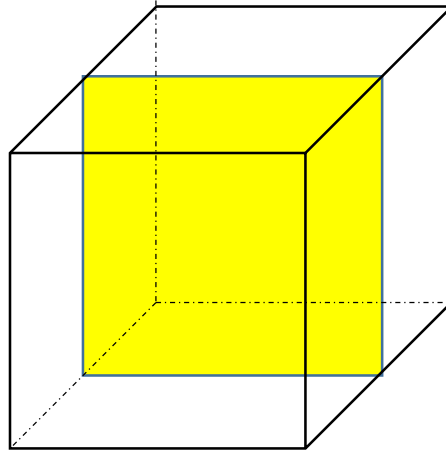
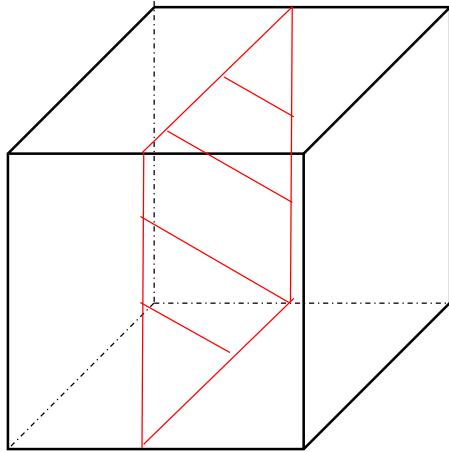
CLASSIFICATION

Crystal System	Conventional Unit Cell	Defining Symmetry
Cubic	$a = b = c$ $\alpha = \beta = \gamma = 90^\circ$	Four 3-fold
Tetragonal	$a = b \neq c$ $\alpha = \beta = \gamma = 90^\circ$	A single 4-fold
Orthorhombic	$a \neq b \neq c$ $\alpha = \beta = \gamma = 90^\circ$	Three orthogonal 2-fold
Monoclinic	$a \neq b \neq c$ $\alpha = \gamma = 90^\circ, \beta \neq 90^\circ$	A single 2-fold
Rhombohedral	$a = b = c$ $\alpha = \beta = \gamma \neq 90^\circ$	A single 3-fold
Hexagonal	$a = b \neq c$ $\alpha = \beta = 90^\circ, \gamma = 120^\circ$	A 6-fold
Triclinic	$a \neq b \neq c$ $\alpha \neq \beta \neq \gamma$	1-fold

Symmetry Element: **Plane of Symmetry**

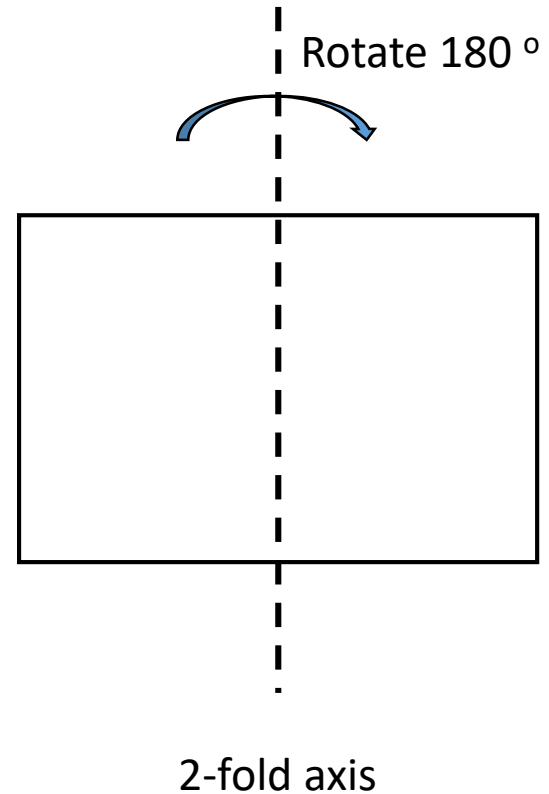
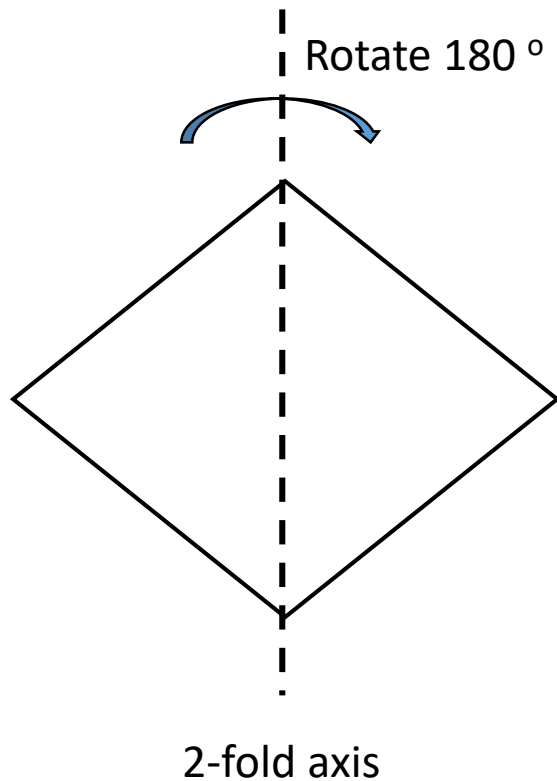
It is an imaginary plane that divides the crystal into two parts in such a way that one part is mirror image of the other part

If an imaginary plane divides the crystal into two parts in such a way that one part is mirror image of the other part, then the crystal is said to have plane of symmetry.



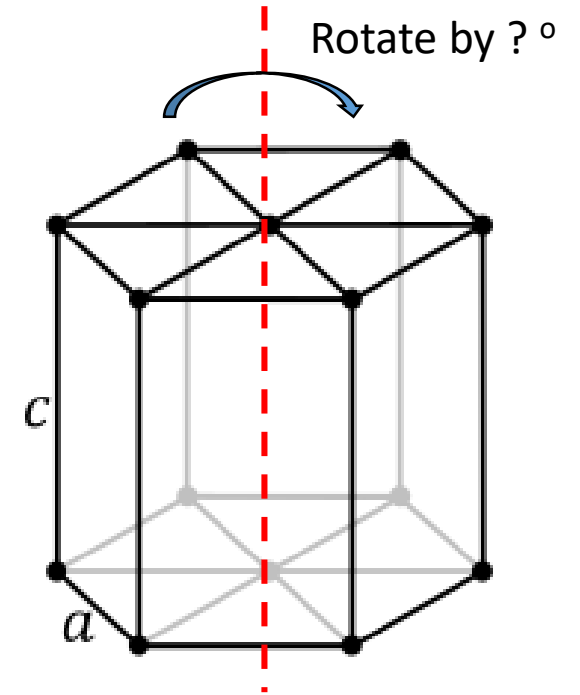
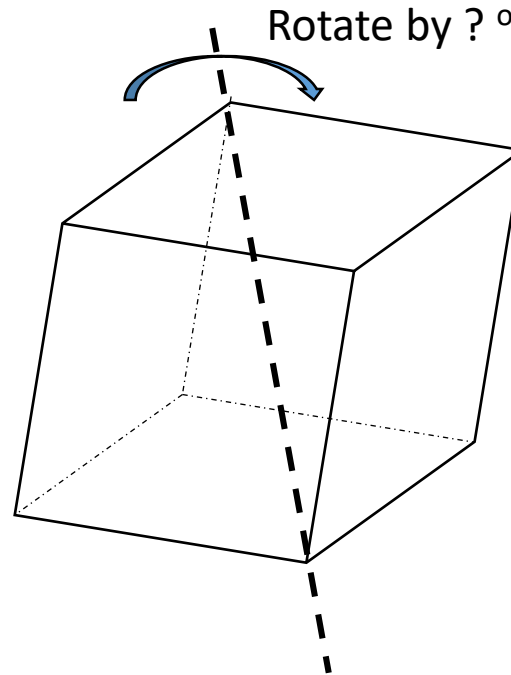
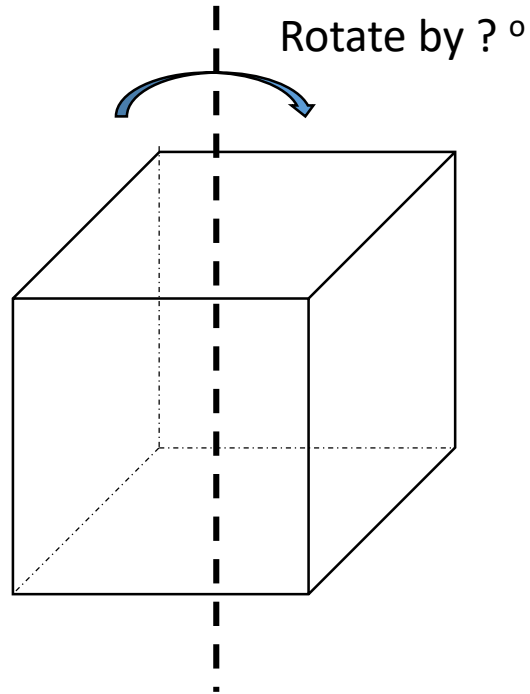
Symmetry Element: **Axis of Symmetry**

It is an imaginary line about which the crystal can be rotated such that its appearance repeated more than once during complete rotation.

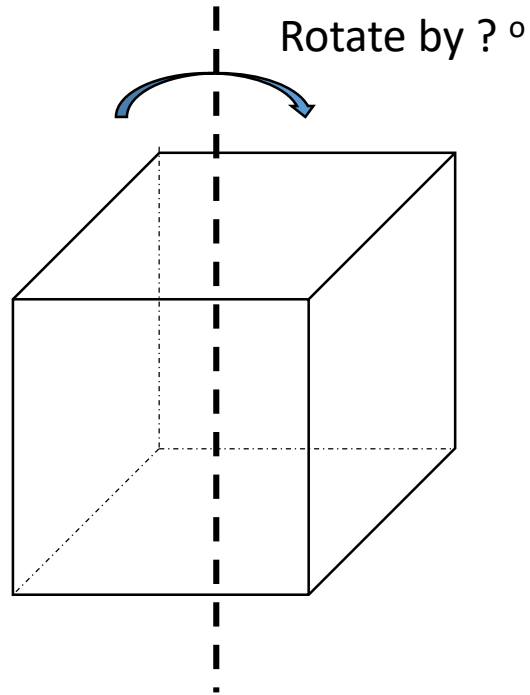


Symmetry Element: **Axis of Symmetry**

It is an imaginary line about which the crystal can be rotated such that its appearance repeated more than once during complete rotation.

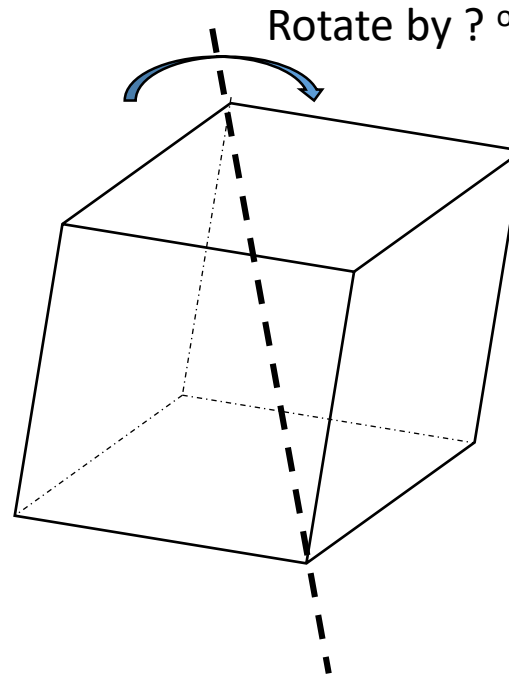


Symmetry Element: **Axis of Symmetry**



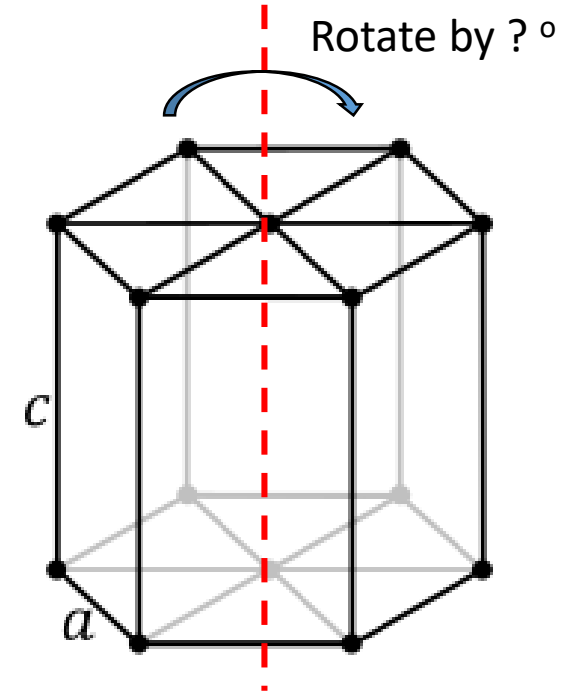
90°

4-fold axis



120°

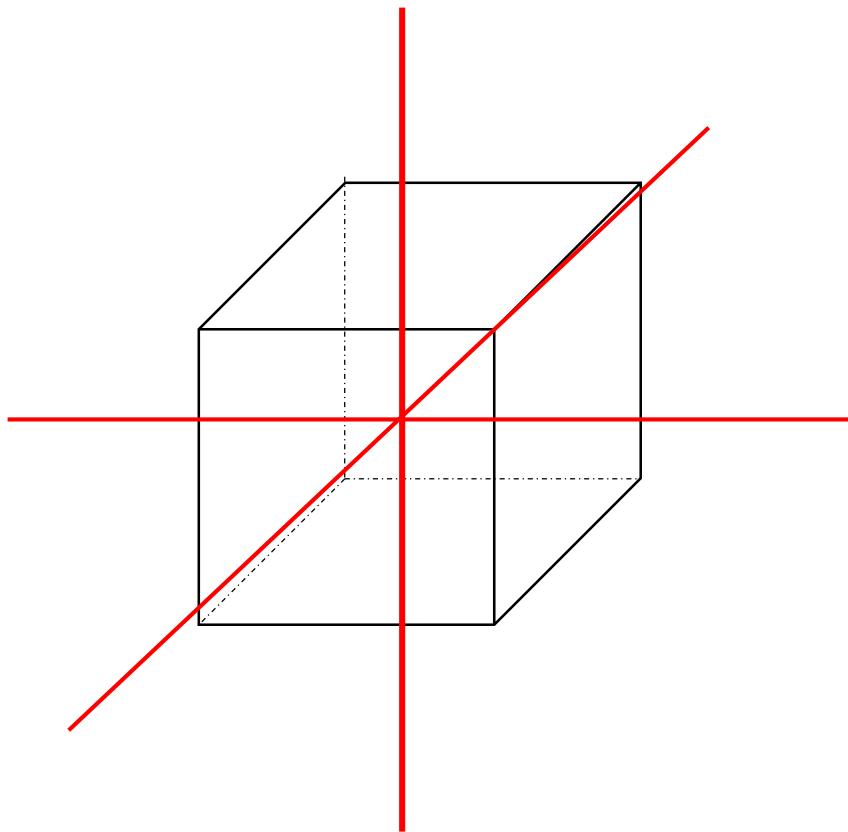
3-fold axis



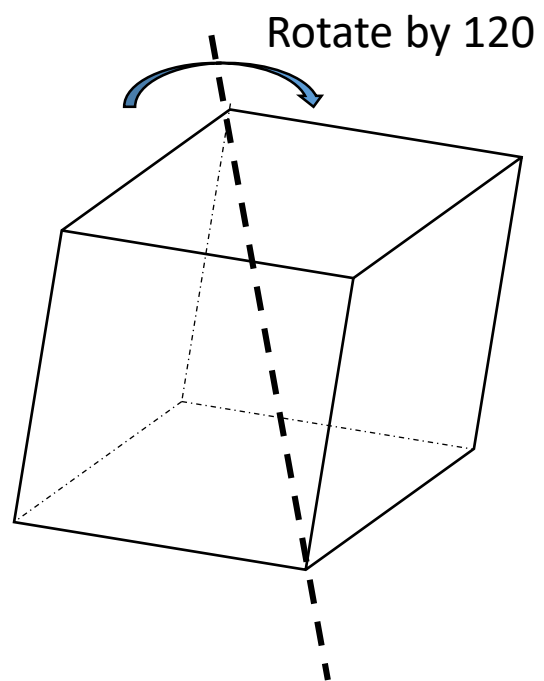
60°

6-fold axis

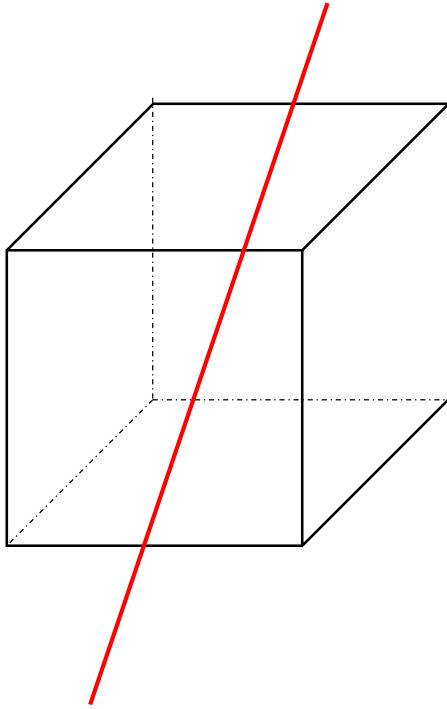
n-fold symmetry: If similar appearance of a crystal is repeated after $360/n$ angle, then the axis is called n-fold axis



Three 4-fold axis



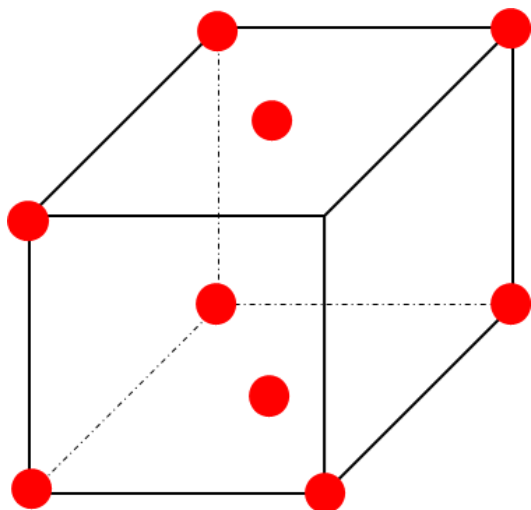
Four 3-fold axis



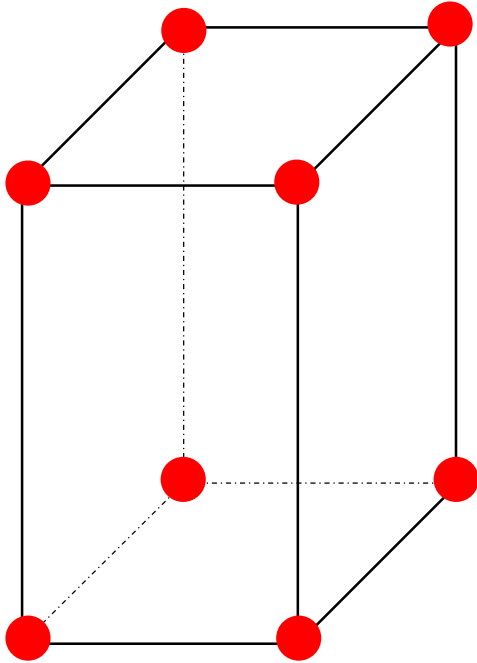
Six 2-fold axis

The defining symmetry of cubic crystal is four 3-fold axis

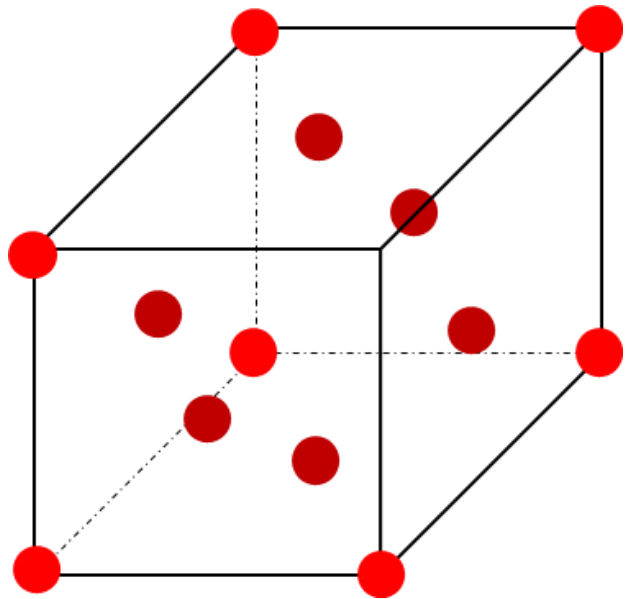
Does cubic C has four 3- fold symmetry?



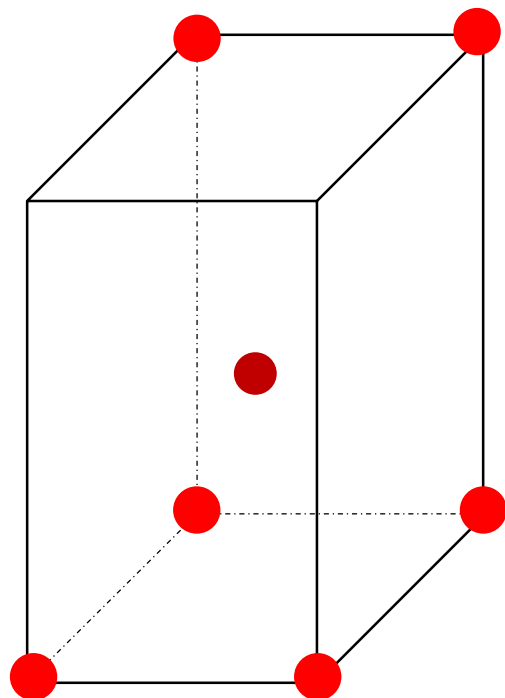
Does tetragonal P has a single 4- fold symmetry?



Does cubic F has four 3- fold symmetry?

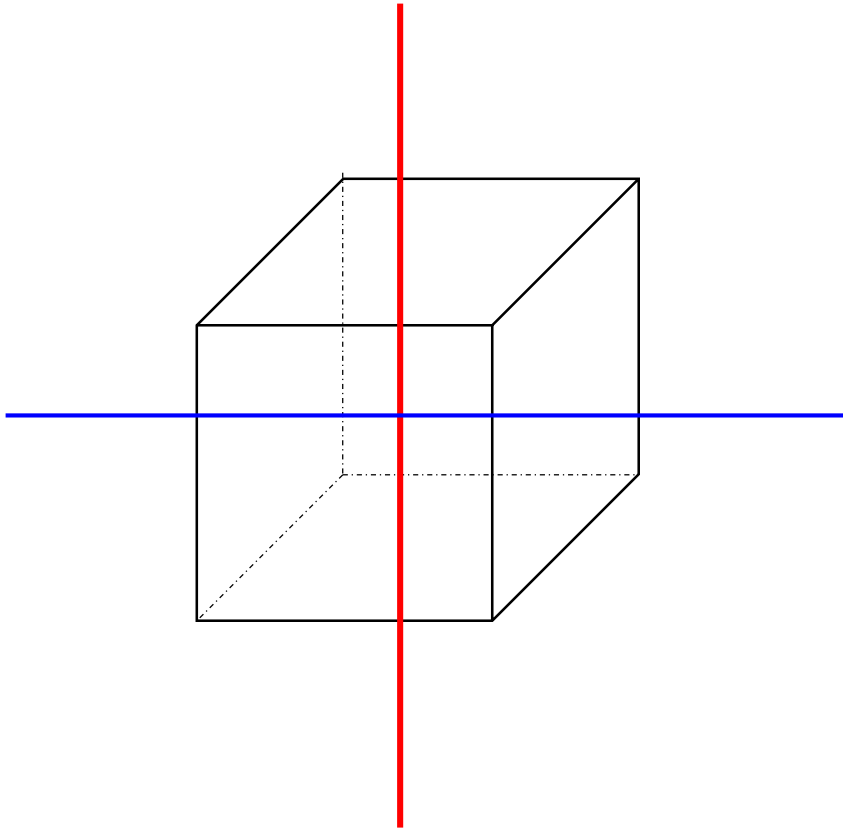


Does tetragonal I has a single 4- fold symmetry?



Centre of symmetry

Centre of symmetry is such a point that any line drawn through it touches the surface of a crystal at equal distance in both direction.

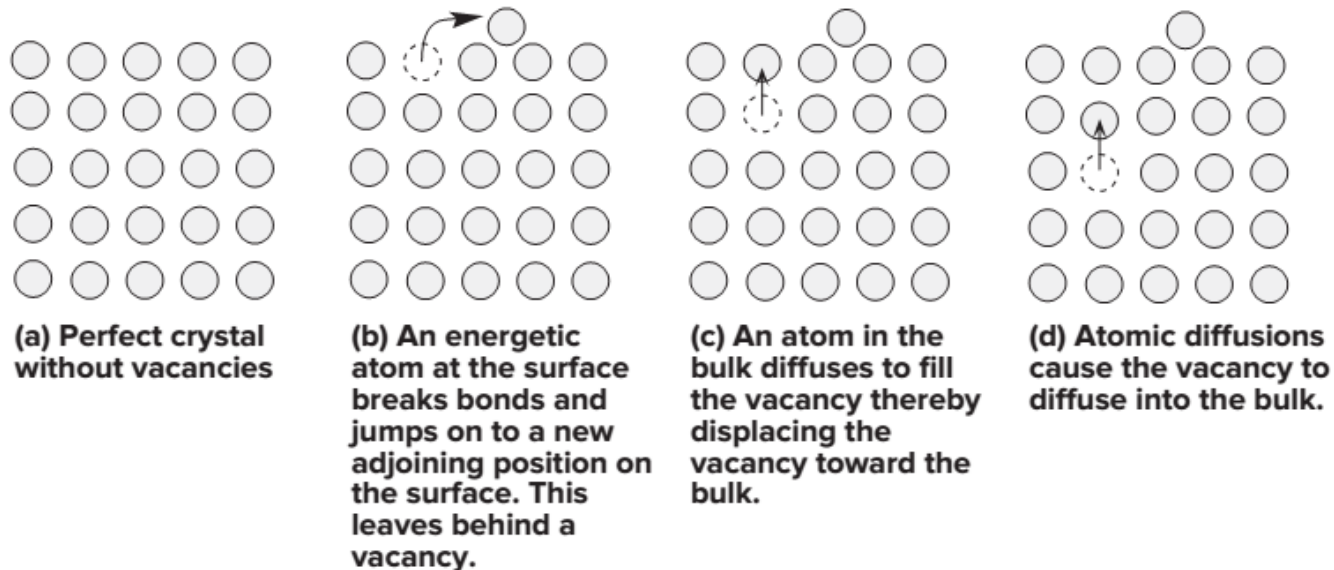


Any crystal has only one center of symmetry.

Symmetry Operation

POINT DEFECTS: VACANCIES AND IMPURITIES

- Above the absolute zero temperature, all crystals have atomic vacancies or atoms missing from lattice sites in the crystal structure. The vacancies exist as a requirement of thermal equilibrium and are called thermodynamic defects.
- All the atoms in a crystal vibrate about their equilibrium positions with a distribution of energies. At some instant, there may be one atom with sufficient energy to break its bonds and jump to an adjoining site on the surface. This leaves a vacancy behind, just below the surface. This vacancy can then diffuse into the bulk of the crystal, because a neighboring atom can diffuse into it.



POINT DEFECTS: VACANCIES AND IMPURITIES

- If E_v is the average energy required to create a vacancy, then only a fraction, $\exp(-E_v/kT)$, of all the atoms in the crystal can have sufficient energy to create vacancies.
- If the number of atoms per unit volume in the crystal is N , then the vacancy concentration n_v is given by

$$n_v = N \exp\left(-\frac{E_v}{kT}\right)$$

VACANCY CONCENTRATION IN A METAL The energy of formation of a vacancy in the aluminum crystal is about 0.70 eV. Calculate the fractional concentration of vacancies in Al at room temperature, 300 K, and very close to its melting temperature 660 °C. What is the vacancy concentration at 660 °C given that the atomic concentration in Al is about $6.0 \times 10^{22} \text{ cm}^{-3}$?

SOLUTION

Using Equation 1.42, the fractional concentration of vacancies are as follows:

At 300 K,

$$\begin{aligned}\frac{n_v}{N} &= \exp\left(-\frac{E_v}{kT}\right) = \exp\left[-\frac{(0.70 \text{ eV})(1.6 \times 10^{-19} \text{ J eV}^{-1})}{(1.38 \times 10^{-23} \text{ J K}^{-1})(300 \text{ K})}\right] \\ &= 1.7 \times 10^{-12}\end{aligned}$$

At 660 °C or 933 K,

$$\begin{aligned}\frac{n_v}{N} &= \exp\left(-\frac{E_v}{kT}\right) = \exp\left[-\frac{(0.70 \text{ eV})(1.6 \times 10^{-19} \text{ J eV}^{-1})}{(1.38 \times 10^{-23} \text{ J K}^{-1})(933 \text{ K})}\right] \\ &= 1.7 \times 10^{-4}\end{aligned}$$

That is, almost 1 in 6000 atomic sites is a vacancy. The atomic concentration N in Al is about $6.0 \times 10^{22} \text{ cm}^{-3}$, which means that the vacancy concentration n_v at 660 °C is

$$n_v = (6.0 \times 10^{22} \text{ cm}^{-3})(1.7 \times 10^{-4}) = 1.0 \times 10^{19} \text{ cm}^{-3}$$

VACANCY CONCENTRATION IN A SEMICONDUCTOR The energy of vacancy formation in the Ge crystal is about 2.2 eV. Calculate the fractional concentration of vacancies in Ge at 938 °C, just below its melting temperature. What is the vacancy concentration given that the atomic mass M_{at} and density ρ of Ge are 72.64 g mol⁻¹ and 5.32 g cm⁻³, respectively? Neglect the change in the density with temperature which is small compared with other approximations in Equation 1.42.

SOLUTION

Using Equation 1.42, the fractional concentration of vacancies at 938 °C or 1211 K is

$$\frac{n_v}{N} = \exp\left(-\frac{E_v}{kT}\right) = \exp\left[-\frac{(2.2 \text{ eV})(1.6 \times 10^{-19} \text{ J eV}^{-1})}{(1.38 \times 10^{-23} \text{ J K}^{-1})(1211 \text{ K})}\right] = 7.0 \times 10^{-10}$$

which is orders of magnitude less than that for Al at its melting temperature in Example 1.18; vacancies in covalent crystals cost much more energy than those in metals.

The number of Ge atoms per unit volume is

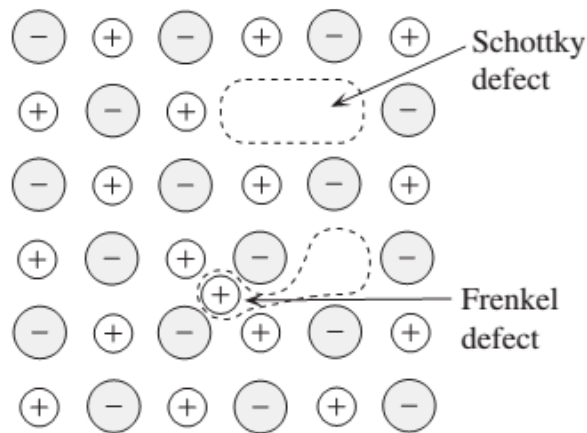
$$N = \frac{\rho N_A}{M_{\text{at}}} = \frac{(5.32 \text{ g cm}^{-3})(6.022 \times 10^{23} \text{ g mol}^{-1})}{72.64 \text{ g mol}^{-1}} = 4.41 \times 10^{22} \text{ cm}^{-3}$$

so that at 938 °C,

$$n_v = (4.4 \times 10^{22} \text{ cm}^{-3})(7.0 \times 10^{-10}) = 3.1 \times 10^{13} \text{ cm}^{-3}$$

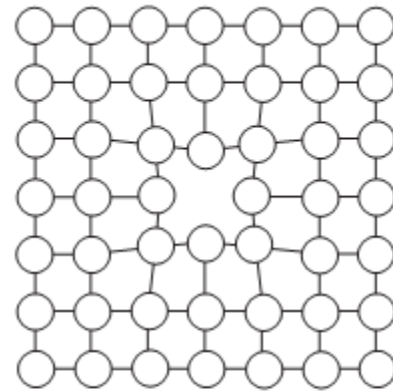
Only 1 in 10⁹ atoms is a vacancy. A better calculation would also consider the decrease in the atomic concentration N with temperature (due to the expansion of the crystal). The final n_v is still about $3 \times 10^{13} \text{ cm}^{-3}$.

Point defects in the crystal structure.

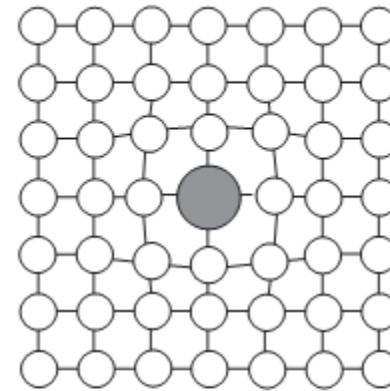


(a) Schottky and Frenkel defects in an ionic crystal.

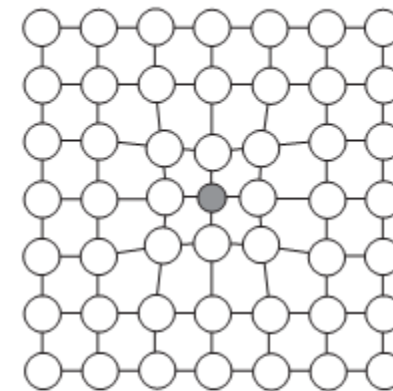
Figure 1.47 Point defects in ionic crystals.



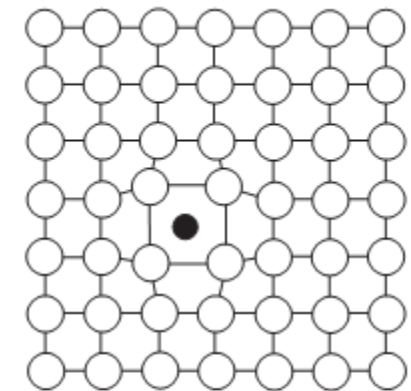
(a) A vacancy in the crystal.



(b) A substitutional impurity in the crystal. The impurity atom is larger than the host atom.



(c) A substitutional impurity in the crystal. The impurity atom is smaller than the host atom.



(d) An interstitial impurity in the crystal. The impurity occupies an empty space between host atoms.

DISLOCATIONS—LINEAR DEFECTS

- A *dislocation* is a linear or one-dimensional defect around which some of the atoms are misaligned.
- One type of dislocation is represented in Figure; an extra portion of a plane of atoms, or half-plane, the edge of which terminates within the crystal.
- This is termed an **edge dislocation**; it is a linear defect that centers on the line that is defined along the end of the extra half-plane of atoms.
- This is sometimes termed the **dislocation line**, which, for the edge dislocation in is perpendicular to the plane of the page.

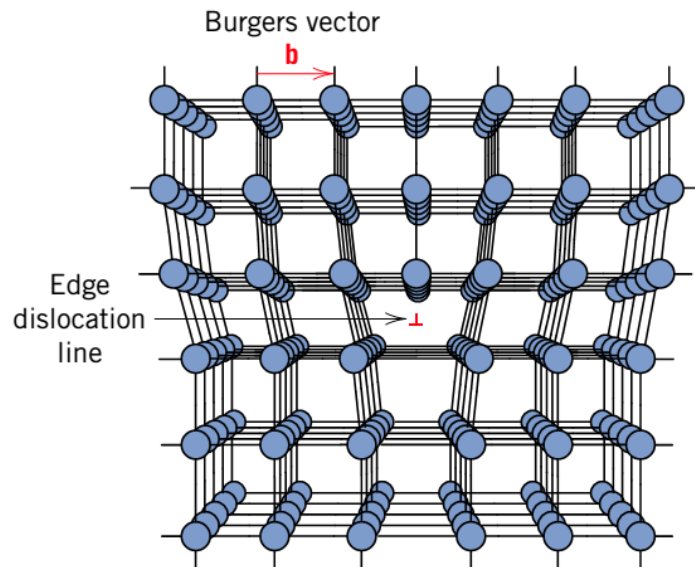
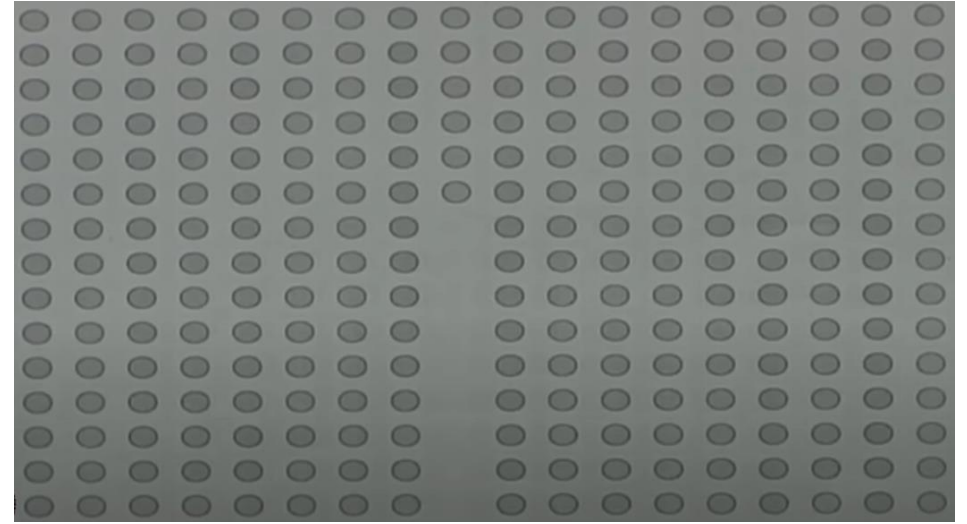
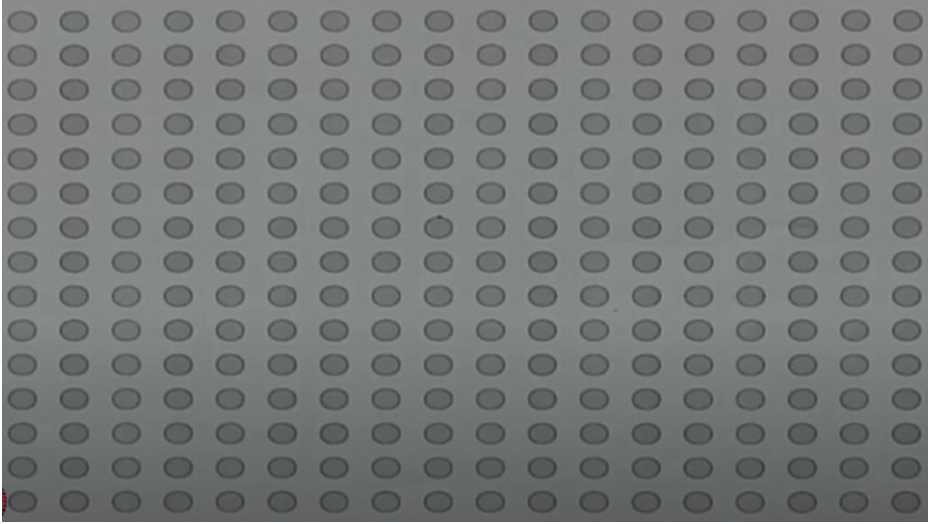
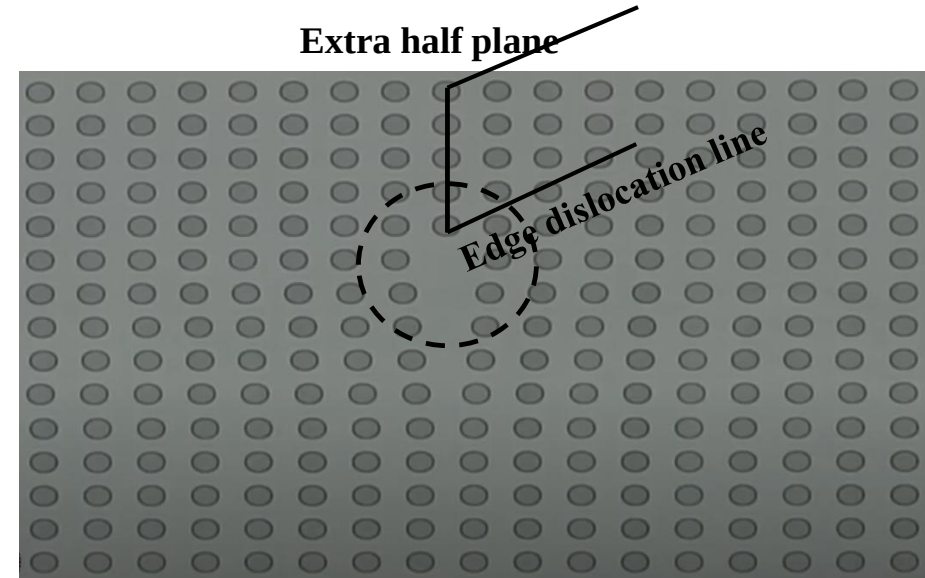
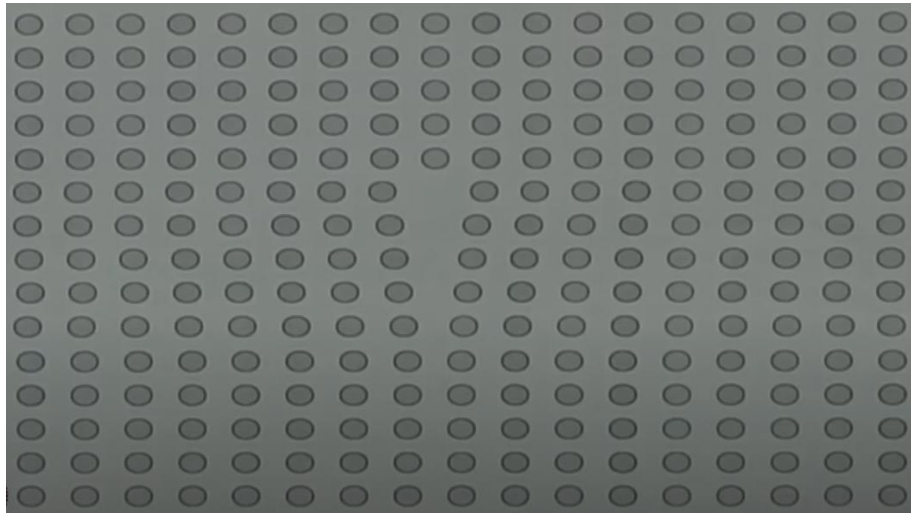


Fig: The atom positions around an edge dislocation; extra half-plane of atoms shown in perspective.

Line Defects: Edge dislocation, half plane



- All the bonds get stretched, and they become unstable.
- Only the bottom edge of the half plane is defect, not the entire plane.



Extra half plane

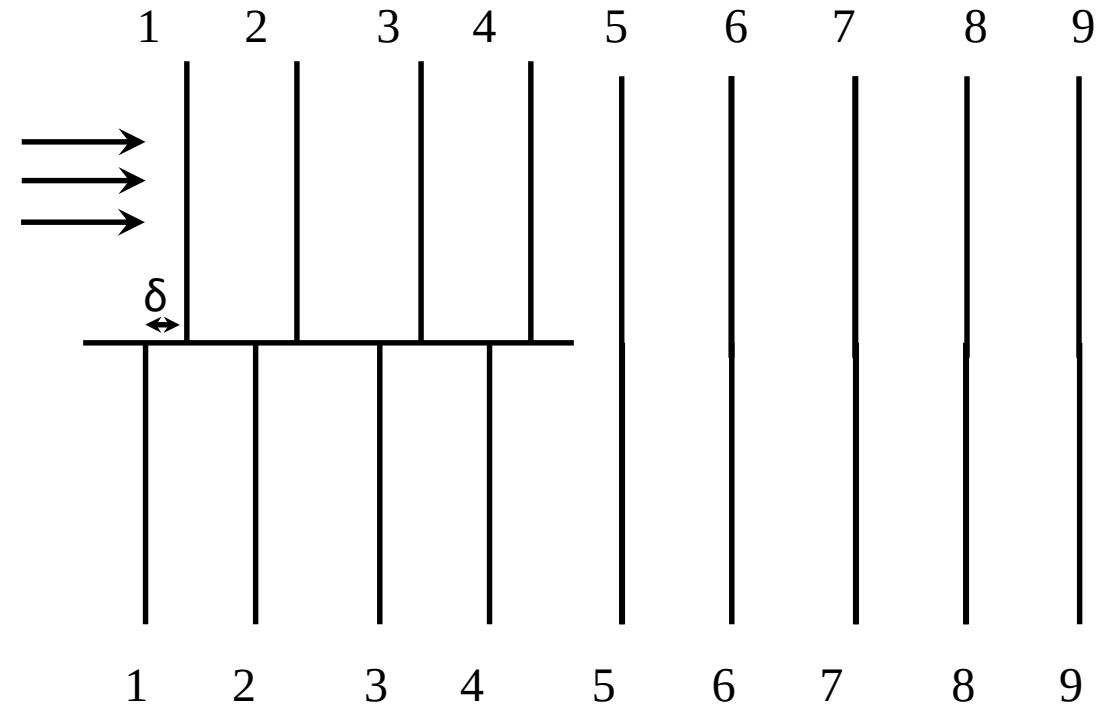
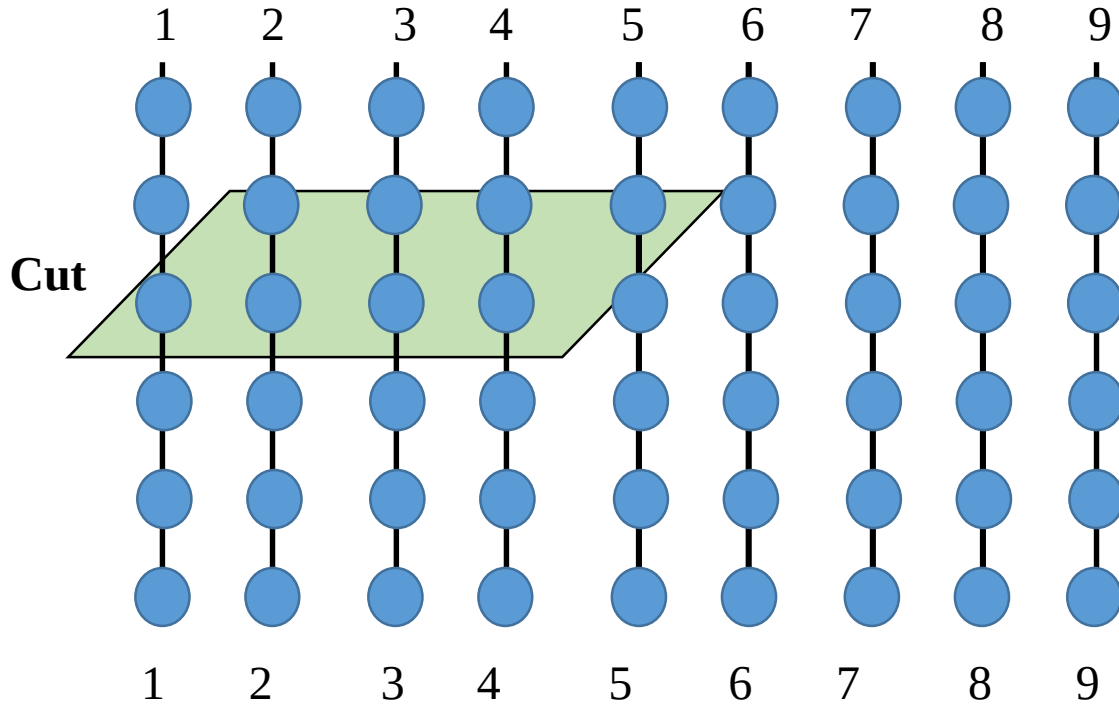
Edge dislocation line

Missing half plane

Line Defects

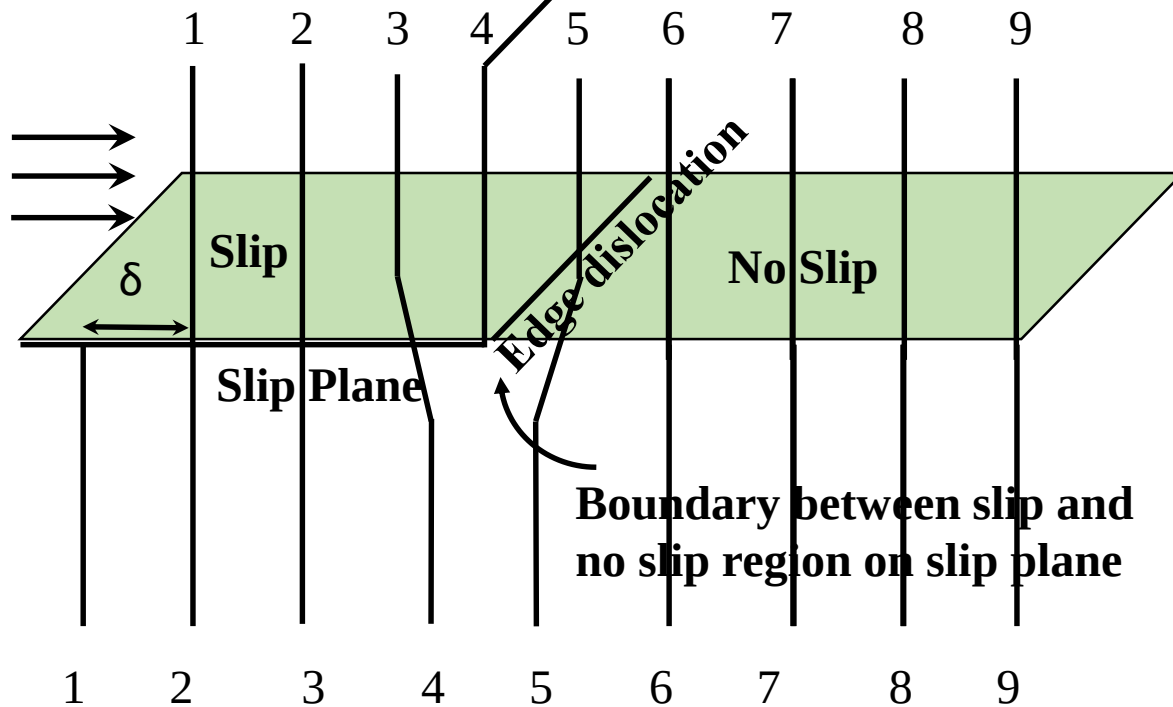
- A line defect is formed in a crystal when an atomic plane terminates within the crystal instead of passing all the way to the end of the crystal.
- The energy required to create a dislocation is typically 100 eV per nm of dislocation line. On the other hand, it takes only a few eV to form a point defect which is a few nm in dimension.
- A line going from the front of the crystal into inside parallel to the bottom edge of the half plane is dislocation line.

Edge dislocation: Slip



Push the upper half of the crystal w.r.t the lower half

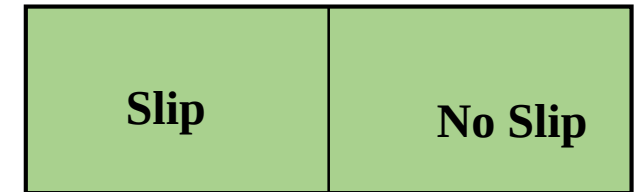
Edge dislocation: Slip



Magnitude and
direction of slip



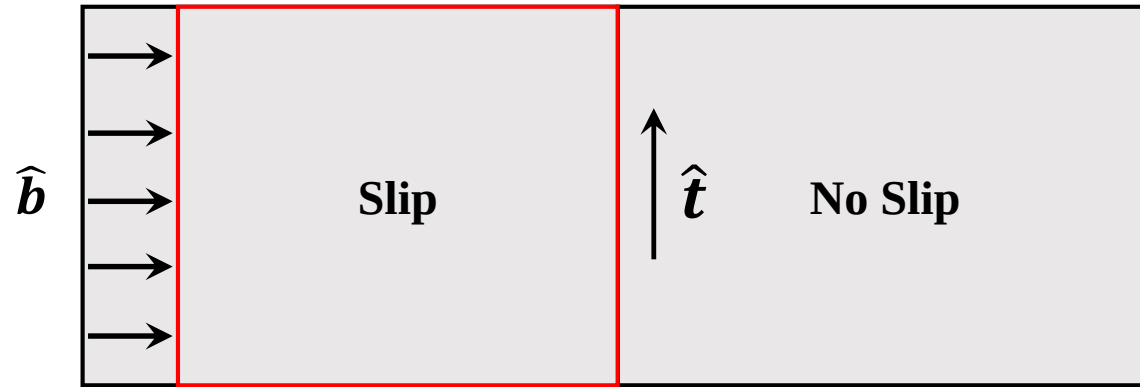
Slip Plane



- A dislocation line is Boundary between slip and no slip region on a slip plane. The orientation of dislocation line is represented by tangent vector ($\hat{t}$)
- The magnitude and direction of slip is known as Burgers vector ($\hat{b}$).

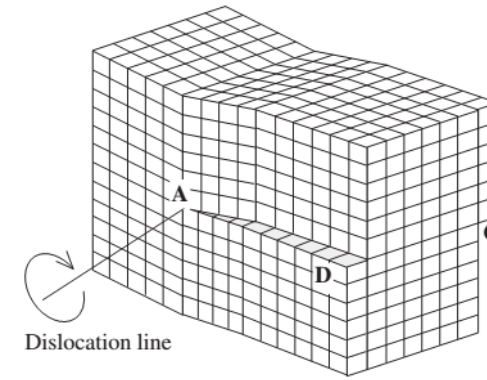
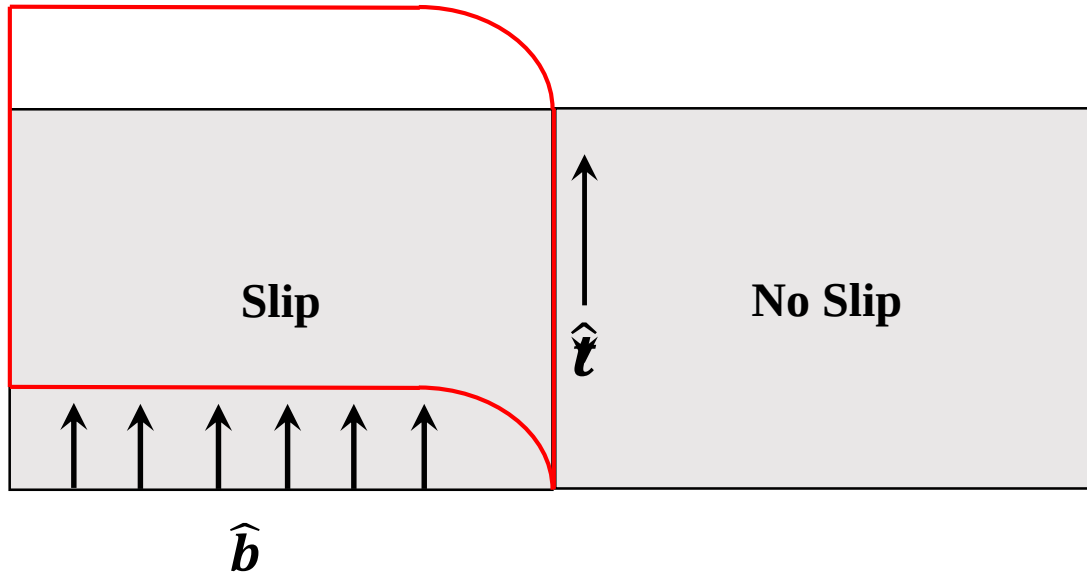
Edge dislocation

- When Burger's vector ($\hat{b}$) is perpendicular to the tangent vector ($\hat{t}$), edge dislocation is formed.

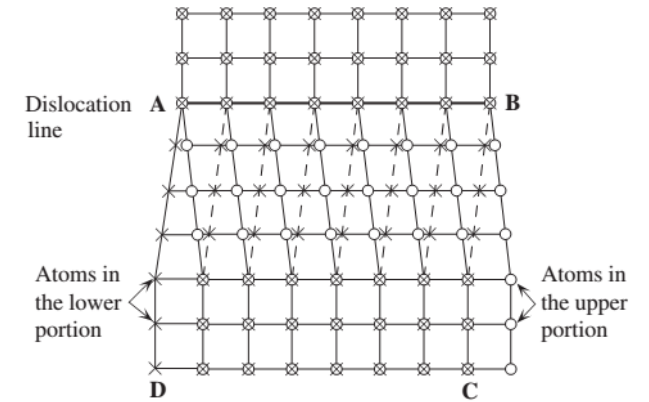


Screw dislocation

- When Burger's vector ($\hat{b}$) is parallel to the tangent vector ($\hat{t}$), screw dislocation is formed.



(a) A screw dislocation in a crystal



(b) The screw dislocation in (a) as viewed from above

Mixed dislocation

- When Burger's vector ($\hat{b}$) and tangent vector ($\hat{t}$) are neither parallel nor perpendicular to each other, mixed dislocation is formed.

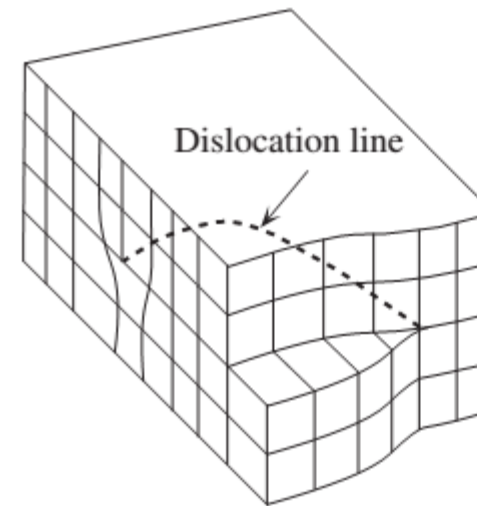
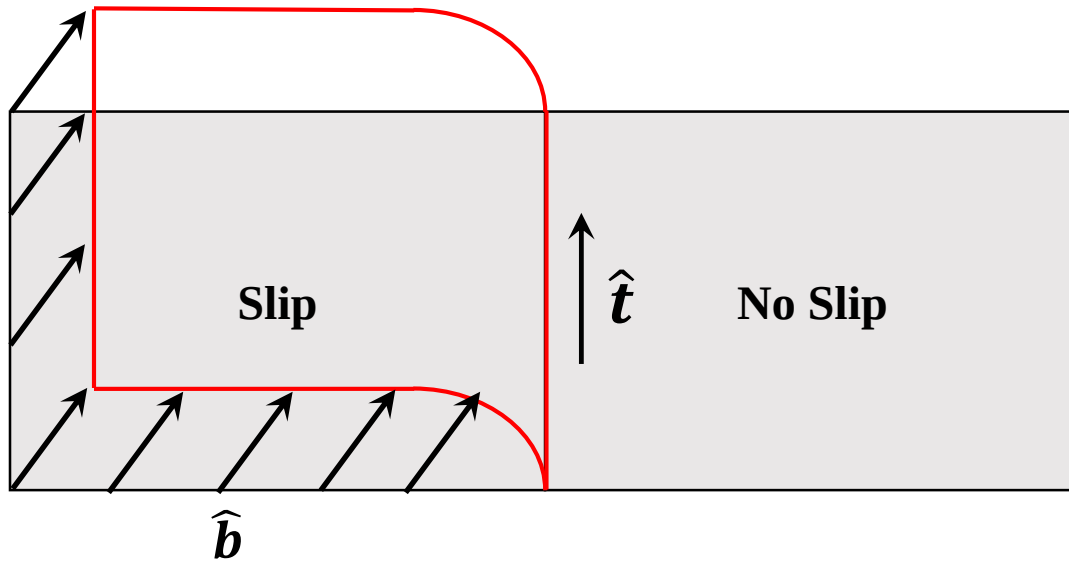


Figure 1.50 A mixed dislocation.

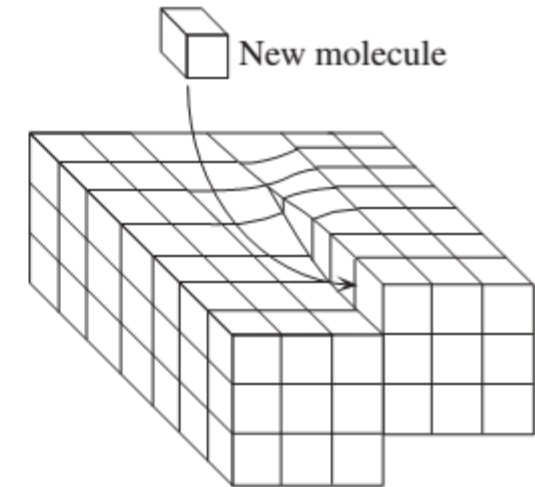


Figure 1.51 Screw dislocation aids crystal growth because the newly arriving atom can attach to two or three atoms instead of one atom and thereby form more bonds.

Grain Boundaries

- Another interfacial defect, the grain boundary, was introduced as the boundary separating two small grains or crystals having different crystallographic orientations in polycrystalline materials.
- A grain boundary is represented schematically in Figure
- Within the boundary region, which is probably just several atom distances wide, there is some atomic mismatch in a transition from the crystalline orientation of one grain to that of an adjacent one.
-

